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A Generalist Biomedical Vision-Language Model via Multi-CLIP Knowledge Distillation

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ABSTRACT

Contrastive Language-Image Pretraining (CLIP) models, which are pretrained on natural images with billions of image-text pairs, exhibit strong zero-shot and cross-modal capabilities. However, their application in biomedicine remains challenging due to limited large-scale image-text data and heterogeneous imaging modalities. Here we show that a generalist biomedical foundation model can be effectively built via multimodal medical knowledge distillation. We introduce MMKD-CLIP, which integrates complementary knowledge from nine biomedical CLIP models. Our two-stage pipeline combines CLIP-style pretraining on 2.9 million biomedical image-text pairs across 26 modalities with large-scale feature-level distillation. We evaluate MMKD-CLIP on 58 datasets spanning nine modalities and six tasks, including classification, retrieval, visual question answering, survival prediction, and cancer diagnosis. MMKD-CLIP performs favorably relative to the teacher models, with results supporting its robustness and cross-domain generalization.

Introduction

Foundation models trained with large-scale vision-language data have advanced general-domain applications, including zero-shot classification, cross-modal retrieval, and open-ended visual question answering (VQA)¹⁻⁵. In parallel, the medical AI community has adapted these foundation models into domain-specific Contrastive Language-Image Pretraining (CLIP)-style² architectures to learn visual-semantic correspondences across radiology⁶⁻⁸, pathology^{9,10}, ophthalmology^{11,12}, and other medical specialties. These biomedical vision-language models (VLMs) have shown utility in tasks such as disease classification, outcome prediction, and visual question answering^{4,13}. However, current biomedical CLIP models are usually designed for specific medical domains. As a result, these models often show limited generalization when applied to data from other medical specialties or different image modalities¹⁴.

Real-world clinical reasoning rarely confines itself to isolated specialties⁴. Physicians integrate multimodal information from radiographs, microscopy, endoscopy, molecular imaging, and clinical notes and guidelines¹⁵. This interdisciplinary nature of medicine suggests the value of a generalist biomedical foundation model capable of unifying visual and textual concepts across specialties. However, building such a model requires large-scale multimodal datasets from diverse sources, a task hindered by inconsistent data standards across institutions and the fragmented nature of biomedical image-text corpora¹⁶. Although several efforts have collected 1.6-6.2 million image-text pairs from PMC-OA¹⁷⁻¹⁹, these numbers remain far below the billions used to train natural image vision-language models²⁰, limiting semantic and modality coverage. In natural image domains, distilling knowledge from a single teacher CLIP model pretrained on billions of pairs into a compact student has yielded competitive performance^{21,22}. Inspired by this, we propose training a student biomedical CLIP model by distilling from multiple domain-specific or generalist biomedical CLIP teachers, each pretrained on millions of pairs, with the aim of improving generalization across biomedical CLIP settings and supporting multimodal integration across medical specialties and imaging modalities.

Here, we present MMKD-CLIP, a generalist biomedical foundation model built through Multi-Modal Medical Knowledge

Distillation with CLIP. As shown in Figure 1 a-c, MMKD-CLIP follows a two-stage pipeline. The first stage performs CLIP-style pretraining on 2,911,190 biomedical image-text pairs spanning 26 imaging modalities from the PMC-OA database. In the second stage, nine biomedical CLIP models are selected as teachers to extract features offline from the same 2.9 million pairs, producing 16,535,905 image-text feature pairs for large-scale knowledge distillation into a single student model. This design allows the student to integrate complementary information from all teachers and improve generalization across diverse modalities. As shown in Figure 1 d and e, we evaluate the model across 6 task types and 9 imaging modalities using 58 datasets, including 38 classification datasets (zero-shot and linear probe), 2 retrieval datasets, 2 VQA datasets, 12 survival prediction datasets, and 4 cancer diagnosis datasets. Across these evaluations, MMKD-CLIP achieves performance comparable to or better than the teacher models. The pipeline can also incorporate future biomedical CLIP models as additional teachers, allowing the student model to be updated as new teacher models become available.

Results

Curating a large-scale biomedical offline distillation dataset

The training pipeline of MMKD-CLIP consists of two stages: pretraining and offline knowledge distillation. In the pretraining stage, we directly used the “Concept-Filtering” subset of the BIOMEDICA dataset¹⁹, which contains about 6 million image-text pairs. To further clean the dataset, we apply an additional filtering step that retains only those pairs annotated with an “image_primary_label” of “Microscopy” or “Clinical Imaging.” In the end, we obtained a total of 2,911,190 image-text pairs, including 1,416,257 single-panel images and 1,494,933 multi-panel images. In Supplementary Table 2, we summarize the distribution of image-text pairs across 26 imaging modalities represented in the filtered subset.

In the offline knowledge distillation stage, we constructed a biomedical offline knowledge distillation dataset containing 16,535,905 (“image”-“text”-“teacher image feature”-“teacher text feature”) quadruplets. The extraction pipeline and rules for these quadruplets are shown in Supplementary Figure 1. This dataset covers 26 imaging modalities and includes feature distributions from 9 biomedical CLIP models. By performing offline knowledge distillation on this dataset, the model can learn joint representations across modalities and CLIP models, which may support generalization across diverse biomedical settings.

Zero-shot biomedical image classification

We first constructed a zero-shot image classification task to evaluate MMKD-CLIP across 9 medical imaging modalities and a total of 38 datasets (Figure 2). Descriptions of these 38 datasets are provided in Supplementary Data 2. The results show that MMKD-CLIP performs competitively across diverse downstream zero-shot image classification tasks, with performance indicating generalization across datasets and modalities. The quantitative results are provided in the Source Data.

MMKD-CLIP attains the highest area under the receiver operating characteristic curve (AUC) among the compared models in 7 of 9 imaging modalities (Magnetic Resonance Imaging (MRI), Fundus, Optical Coherence Tomography (OCT), Computed Tomography (CT), X-ray, Endoscopy, and Pathology), and is ranked among the top 3 models in the remaining 2 modalities. Here, “Rank 1 of 10 in a modality” in Figure 2 refers specifically to the average dataset-level ranking within that modality, rather than indicating that MMKD-CLIP is ranked first on every individual dataset. The corresponding dataset-wise ranking distributions for all modalities are provided in Supplementary Figure 7, which shows the variability and relative ranking of each model across individual datasets.

Specifically, MMKD-CLIP shows AUC improvements in MRI (1.23% v s. the second-best model, $P = 0.176$), fundus imaging (0.24% vs. the second-best model, $P = 0.002$), OCT (0.24% vs. the second-best model, $P = 0.332$), CT (0.90% vs. the second-best model, $P = 0.002$), X-ray (3.25% vs. the second-best model, $P = 0.002$), Endoscopy (0.45% vs. the second-best model, $P = 0.0679$), and Pathology (2.48% vs. the second-best model, $P = 0.002$). In ultrasound and dermatology, where other general biomedical CLIP models perform better on average, MMKD-CLIP remains among the top 3 models. Across individual datasets, MMKD-CLIP achieves the best or second-best performance in 24 of 38 tasks, as shown in the rank distribution plots. These results suggest that MMKD-CLIP benefits from multimodal knowledge distillation and performs competitively across modality-specific diagnostic tasks.

MMKD-CLIP shows more stable performance across imaging modalities than existing biomedical CLIP variants. As shown in Figure 2, MMKD-CLIP often ranks higher at the modality level (averaged across datasets) than at the individual dataset level. For example, MMKD-CLIP ranks 1st in the MRI, CT, Fundus, OCT, Endoscopy, and X-ray modalities, while ranking 2nd or 3rd on some constituent datasets (e.g., BreastTumorMRI or SIIMACR). This pattern indicates that its aggregated performance across datasets is less influenced by any single dataset. In contrast, models such as BiomedCLIP and UniMedCLIP show greater variation between dataset-level and modality-level rankings. For instance, BiomedCLIP ranks 1st on specific datasets such as NCT-CRA (Pathology) or WCE (Endoscopy), but its average rank is lower at the modality level. Similarly, UniMedCLIP achieves a top-2 ranking on OCTMNIST but does not maintain the same relative ranking across the broader OCT modality. Overall, MMKD-CLIP maintains relatively consistent performance across diverse imaging modalities, a property that may be relevant for applications involving variation across institutions, devices, and patient populations.

Linear probing image classification

We compare MMKD-CLIP with nine existing biomedical foundation models, including MedCLIP, PLIP, PUBMEDCLIP, BMCA, GenMedClip, PMCCLIP, BiomedCLIP, UniMedCLIP, and QuiltNet, using linear probing classification tasks across 9 medical imaging modalities (Figure 3 a). Evaluation was performed under three training data regimes (1%, 10%, and 100%) to examine performance under varying levels of supervision. Across these settings, MMKD-CLIP ranks first or is tied for the highest performance in most modality-proportion combinations. The quantitative results are in the Source Data.

In the MRI modality, MMKD-CLIP achieves an AUC of 94.44% at 1% training data (95% CI: 92.55-96.09%), exceeding the second-best model, GenMedClip, by 5.46%. In Fundus, MMKD-CLIP achieves an AUC of 79.76% (95% CI: 77.69-81.78%), exceeding UniMedCLIP by 5.69%. These statistically significant differences in high-resolution structural modalities indicate improved performance in these settings. In pathology, MMKD-CLIP achieves an AUC of 88.61% with 1% training data, compared with 86.25% for PLIP, which was trained specifically for pathology. This result indicates that MMKD-CLIP also performs competitively on pathology images.

At higher data availability (10% and 100%), all models improve, and MMKD-CLIP remains among the top-performing models. For example, in X-ray, the AUC increases from 88.03% (1%) to 91.02% (100%), with MMKD-CLIP ranking above the other models at each training proportion. In fundus imaging, MMKD-CLIP achieves an AUC of 89.95% at 100% training data, compared with 88.59% for the next-best model, with minimal overlap between the confidence intervals (95% CI: 88.56-91.16%). These results indicate a statistically significant difference under the full-training setting.

On average across all modalities at the 1% training level, MMKD-CLIP achieves a macro AUC of 84.51%, representing a 0.37% improvement over the strongest baseline model. Similar differences are observed under both low-data and full-data conditions, suggesting that MMKD-CLIP provides competitive generalization and representation performance across training settings. The consistent performance gap across diverse modalities, from cross-sectional imaging (e.g. MRI and CT) to photographic domains such as dermatology and fundus imaging, indicates that MMKD-CLIP performs competitively across a range of diagnostic imaging contexts. Collectively, these results support MMKD-CLIP as a competitive vision-language foundation model for medical image understanding.

Cross-modal retrieval

To evaluate vision-language alignment, we compare MMKD-CLIP with nine baseline medical foundation models on cross-modal retrieval tasks using two diverse datasets: the MedTrinity-25M subset²³, a heterogeneous clinical multimodal corpus, and BookSet²⁴, an academic captioning dataset derived from ten academic textbooks (Figure 3 b). The tasks include both text-to-image and image-to-text retrieval, with performance measured by Recall@1, Recall@10, and Recall@50. The quantitative results are in the Source Data.

On the MedTrinity-25M subset, MMKD-CLIP shows generally higher retrieval recall than the compared models across most metrics. For text-to-image retrieval, MMKD-CLIP achieves a Recall@50 of 15.16% (95% CI: 14.15-16.11%), compared with 13.67% for the second-best model, BMCA (95% CI: 12.73-14.66%), corresponding to a difference of 1.49% ($P = 0.002$). MMKD-CLIP achieves a Recall@10 of 3.76%, compared with 3.74% for UniMedCLIP ($P = 0.117$). MMKD-CLIP also achieves a Recall@1 of 0.68%, whereas most baseline models remain below 0.5% ($P = 0.002$). These results suggest that MMKD-CLIP performs well on matching text and images in a heterogeneous medical dataset. In the image-to-text retrieval task on the MedTrinity-25M subset, MMKD-CLIP achieves the highest Recall@50 (16.63%, 95% CI: 15.62-17.72%), compared with 15.06% for BiomedCLIP, a difference of 1.57% ($P = 0.002$), and achieves similar performance at Recall@10 (4.20%) and Recall@1 (0.38%, $P = 0.002$). Although some competing models (e.g., BMCA) show similar Recall@10 values (4.19%, $P = 0.547$), no statistically significant difference was observed between MMKD-CLIP and BMCA at this threshold. Nevertheless, MMKD-CLIP maintains competitive and relatively consistent performance across retrieval directions and recall thresholds.

BookSet contains better-aligned image-text pairs with fewer domain-specific abbreviations and less noise. MMKD-CLIP continues to perform competitively on this dataset. In the text-to-image setting, Recall@50 increases to 52.95% (95% CI: 51.52-54.38%), compared with 52.36% for UniMedCLIP ($P = 0.002$). For image-to-text retrieval, MMKD-CLIP achieves a Recall@10 of 26.61%, corresponding to a difference of 0.18% relative to UniMedCLIP ($P = 0.017$), although the absolute difference was small. MMKD-CLIP also achieves the highest Recall@1 of 8.45%, indicating competitive performance in fine-grained image-text matching.

Collectively, these results indicate that MMKD-CLIP performs competitively in retrieval on both noisy, large-scale clinical data (MedTrinity) and more curated academic datasets (BookSet). The observed differences across both retrieval directions, particularly at lower recall thresholds (e.g., Recall@1), suggest improved multimodal alignment relative to existing medical CLIP variants. These findings support the use of MMKD-CLIP as a foundation model for downstream multimodal medical tasks.

Visual question answering

To evaluate the visual question answering performance of MMKD-CLIP, we conduct evaluations on two established VQA benchmarks. The first is SLAKE²⁵, which focuses on multi-organ CT-based anatomical and clinical understanding. The second is VQA-RAD²⁶, a manually curated dataset consisting of 3,515 question-answer pairs, in which medical professionals pose natural language questions about radiological images and provide corresponding reference answers. The tasks are categorized into open-ended, closed-form, and overall questions, with accuracy as the main evaluation metric. The quantitative results are presented in the Source Data.

As shown in Figure 4 a, MMKD-CLIP achieves the highest accuracy among the compared models across both datasets and question types. On SLAKE, MMKD-CLIP achieves the highest accuracy on overall questions (80.21%, 95% CI: 77.95-82.56%), compared with 79.83% for the next-best model, PMCCLIP ($P = 0.007$), and 79.26% for MedCLIP ($P = 0.002$). In closed questions, MMKD-CLIP attains an accuracy of 83.65%, corresponding to a 0.72% difference relative to PMCCLIP. On VQA-RAD, MMKD-CLIP also ranks highest. For closed questions, MMKD-CLIP achieves 79.34%, the highest value among the evaluated models, and an overall accuracy of 70.07%. These results indicate competitive performance across both datasets and question types.

To further examine model predictions, we present qualitative visualizations of selected VQA samples (Figure 4 b & c). In SLAKE (Figure b), MMKD-CLIP correctly identifies anatomical structures (e.g., “liver” on the left side) and spatial relations (e.g., “small bowel on the right”) in the examples shown. In contrast, models such as MedCLIP and GenMedClip produce less accurate responses on some samples, particularly for open-ended questions. In VQA-RAD (Figure c), MMKD-CLIP correctly identifies locations (e.g., “anterior mediastinum”) and diagnostic findings (e.g., “calcified atherosclerosis”) in the illustrated cases. While BMCA and UniMedCLIP also provide correct answers for many samples, they produce less accurate responses in some cases involving ambiguous wording or visual occlusion (e.g., “sulcal abnormalities”).

Collectively, these results show that MMKD-CLIP achieves competitive VQA performance across the evaluated settings, including detailed visual recognition and medical question answering. It performs competitively across both quantitative metrics and qualitative examples. These results support its evaluation on multimodal medical tasks.

Survival prediction task

To further evaluate MMKD-CLIP on multimodal survival prediction, we analyze its performance across 12 distinct cancer types from the TCGA dataset²⁷, incorporating pathology patch images, reports, and their fusion. Kaplan-Meier (KM) curves and concordance indices (C-Index) are used to evaluate risk stratification and predictive performance, respectively. The quantitative results are in the Source Data.

As visualized in Figure 5 a, patients were classified into high-risk and low-risk groups based on model-predicted hazard. Across most cohorts, stratification by MMKD-CLIP is associated with statistically significant separation between the survival curves. For instance, in BRCA (N=138), the survival curves diverge significantly ($P = 0.000752$), with a hazard ratio (HR) of 8.22. Similar patterns are observed in LUAD (HR = 6.44, $P = 0.000528$), HNSC (HR = 5.99, $P = 0.000007$), and PAAD (HR = 26.22, $P = 0.000016$). While small datasets such as MESO (N=11) show greater variance, the model also separates risk groups in this cohort (HR = 15.64, $P = 0.001972$).

We further compare survival prediction performance using the C-Index metric between MMKD-CLIP and PLIP across unimodal and multimodal inputs (Figure 5 b). Overall, MMKD-CLIP achieves an average C-Index of 0.6873 using both image and report inputs, compared with 0.6675 for PLIP. Across multiple cancers, including STAD, KIRC, and BLCA, the combination of image and report inputs is associated with higher C-Index values than either modality alone. For example, in CESC, MMKD-CLIP achieves a C-Index of 0.7652 with multimodal input, compared with 0.6643 for image-only input and 0.6712 for report-only input.

MMKD-CLIP achieves higher C-Index values than PLIP across both common cancers (e.g., BRCA and LUAD) and lower-sample-size cancers (e.g., SARC and MESO). The standard deviation bars are also smaller for MMKD-CLIP in several settings, particularly under multimodal fusion. These results show that MMKD-CLIP performs competitively for survival risk prediction when visual and textual inputs are modeled jointly.

Supervised cancer diagnosis task

To evaluate MMKD-CLIP on supervised cancer classification tasks after fine-tuning, we conduct experiments on four benchmark datasets spanning multiple imaging modalities and anatomical domains: BRACS-3²⁸, BRACS-7²⁸, TCGA-NSCLC²⁹, and Camelyon16³⁰. As shown in Figure 6 a, AUC is used to assess classification performance. The quantitative results are presented in the Source Data.

On BRACS and TCGA-NSCLC, MMKD-CLIP achieves higher AUC values than the compared general biomedical models (e.g., BMCA and GenMedClip) and domain-specific models (e.g., QuiltNet and PLIP). On BRACS-3, MMKD-CLIP achieves an AUC of 87.42%, compared with 86.44% for the best baseline, PLIP ($P = 0.008$). A similar pattern is observed for BRACS-7, a 7-class variant, where MMKD-CLIP reaches an AUC of 88.13%. For TCGA-NSCLC, MMKD-CLIP achieves an AUC

of 97.28%, which is higher than the compared multimodal and unimodal pretrained models ($P = 0.013$). On Camelyon16, MMKD-CLIP achieves an AUC of 89.17%, which is higher than that of most compared models, including QuiltNet and PLIP ($P = 0.002$).

These results show that MMKD-CLIP performs competitively across different cancer classification tasks, including radiology and pathology, binary and multi-class classification, and coarse and more detailed label settings. The observed performance differences across datasets are consistent with the use of multi-teacher distillation and unified vision-language pretraining in this model.

Ablation of loss components

We conducted an ablation study on the FD and ICL losses used in distillation (Figure 6 b). Across all domains, removing either FD or ICL is associated with reduced performance. Specifically, removing the FD loss results in a 0.09–1.26% decrease across modalities, whereas removing the ICL loss results in a 0.14–0.72% decrease. The full MMKD-CLIP shows the highest accuracy among the evaluated configurations, with an overall accuracy of 84.78% (95% CI: 82.51–86.91%), compared with 84.69% for the second-best configuration (without FD loss) and 84.26% for the pretraining-only baseline. In the X-ray domain, the model without ICL performs below the version without FD loss (82.52% vs. 82.99%). In contrast, the pathology and CT domains show greater differences after removing the FD loss. The MRI domain shows the highest classification scores overall, with MMKD-CLIP reaching 90.09% (95% CI: 87.61–92.30%). The difference between the pretraining-only model (88.90%) and the full MMKD-CLIP (90.09%) is also observed in this modality.

These results show that both loss components are associated with model performance, and that combining ICL loss with FD loss yields higher zero-shot biomedical classification performance across diverse modalities than the ablated settings.

Fidelity of dimensionality-alignment strategies

Alignment of heterogeneous teacher embeddings into a shared representation space is required for feature-level distillation in our framework. Seven of the nine teacher models employ ViT-B encoders and produce 512-dimensional embeddings, whereas BMCA and PMCCLIP are based on ViT-L encoders with 768-dimensional outputs. Because the contrastive, FD, and ICL losses require features of equal dimensionality, all teacher representations are projected to a common 512D space before distillation. We therefore examined whether this alignment step affects information content or alters the modality-specific geometry of the teacher embeddings. The resulting analyses are summarized in Supplementary Figure 2.

We first assessed the reconstruction error of the shared dual-stream autoencoder used in the AE-Projection strategy. As shown in Supplementary Figure 2a, the mean squared error between the original and reconstructed features is consistently on the order of 10^{-4} for both image and text embeddings across all teacher models. These results indicate that the reconstructed features are close to the original feature representations, with reconstruction errors on the order of 10^{-4} across models.

To evaluate the preservation of geometric structure, we computed the Gram-matrix error between original teacher features and their projected counterparts obtained via PCA reduction, AE projection, and AE reconstruction (Supplementary Figure 2b). For an original feature matrix $\mathbf{X} \in \mathbb{R}^{N \times d}$ and a processed version $\tilde{\mathbf{X}} \in \mathbb{R}^{N \times d'}$, we first map features to the same dimension when necessary and compute their Gram matrices:

$$\mathbf{G} = \mathbf{X}\mathbf{X}^\top, \quad \tilde{\mathbf{G}} = \tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top. \quad (1)$$

The geometric fidelity error is defined as:

$$\text{GramErr}(\mathbf{X}, \tilde{\mathbf{X}}) = \frac{\|\mathbf{G} - \tilde{\mathbf{G}}\|_F}{\|\mathbf{G}\|_F}. \quad (2)$$

Lower values indicate closer preservation of pairwise inner-product structure. Across both image and text branches, AE-Projection yields lower Gram-matrix error than PCA for BMCA and PMCCLIP, the two models in this set that require dimensionality reduction. Compared with AE-Rec, AE-Projection also achieves comparable or lower Gram-matrix error for several models, with the clearest advantage on PMCCLIP. This result suggests that direct projection can preserve the original embedding geometry effectively, and in some cases even more faithfully than reconstruction-based features.

To contextualize these findings, Supplementary Figure 2c-d summarizes the three alignment strategies evaluated: AE-Projection, in which all teacher embeddings are mapped to 512D using CLIP-specific projection encoders; PCA+default, in which only BMCA and PMCCLIP are reduced via PCA while all ViT-B teachers remain unchanged; and AE-Projection+default, in which only BMCA and PMCCLIP undergo AE-based reduction. The latter two serve as controls designed to preserve geometric relationships by minimizing transformations applied to ViT-B-based teachers.

Finally, Supplementary Figure 2e reports zero-shot classification performance across nine imaging modalities under all three strategies. Performance is nearly identical across settings, with AE-Projection matching or slightly outperforming the two control strategies despite applying uniform projections. These results suggest that the representational perturbations introduced by AE-Projection have limited impact on downstream tasks.

Together, these analyses suggest that the dimensionality-alignment step maintains low reconstruction error, preserves modality-specific geometry at a level comparable to or better than PCA, and produces aligned features that maintain performance in downstream tasks. These results support the use of AE-Projection as a unification mechanism for feature-level multi-teacher distillation.

Number and composition of teacher models

To further analyze the influence of teacher composition, we designed three incremental integration strategies based on the number of feature quadruples contributed by each teacher model. As shown in Supplementary Figure 3c, teachers are ordered from left to right by increasing quadruple count; for example, MedCLIP contributes the fewest quadruples, whereas BiomedCLIP contributes the most. In Strategy A, teachers are added from low to high data contribution, simulating the progressive inclusion of larger-scale teachers. In Strategy B, teachers are added from high to low data contribution, testing whether early distillation from data-rich generalists reduces the relative contribution of smaller specialists. In Strategy C, all modality-specific specialist teachers are included first, followed by the progressive addition of generalist teachers (BiomedCLIP, BMCA, GenMedClip and UniMedCLIP).

As illustrated in Supplementary Figure 3a-b, all three strategies show consistent performance gains as the number of teachers increases from one to four, but the improvement saturates beyond six teachers. Strategy C shows a favorable balance between modality coverage and generalization. These results suggest that retaining domain-specific experts while gradually incorporating generalist teachers is associated with complementary knowledge transfer. In contrast, Strategy A shows slower convergence due to the initial reliance on small-scale modality experts, whereas Strategy B shows signs of overfitting to dominant modalities at early stages. These results indicate that the order and heterogeneity of teacher integration appear to play a larger role than the absolute number of teachers, supporting the use of a hybrid ensemble that combines both generalist and specialist biomedical CLIP models.

Effect of generalist and modality-specific teachers

To disentangle the roles of modality-specific and generalist teachers, we performed a controlled ablation study on both X-ray and Pathology benchmarks (Supplementary Figure 4) under four training settings. In the first setting, we trained a baseline CLIP model using only data from the target modality (X-ray or Pathology) with a standard contrastive objective, corresponding to the first-stage MMKD-CLIP model ("MMKD-CLIP Stage 1 Only"). This pure CLIP-style encoder, trained without teacher supervision, was compared against established modality-specific baselines (MedCLIP for X-ray and PLIP or QuiltNet for Pathology). Although the first-stage model achieved comparable performance, it remained below the best specialized teacher in each domain, suggesting that contrastive pretraining on a single modality alone may be insufficient to capture domain-specific semantics fully.

In the second setting, we kept the training data restricted to the target modality and introduced distillation only from modality-specific teachers (for example, MedCLIP for X-ray and PLIP, QuiltNet, or their combination for Pathology). As shown by the blue bars in Supplementary Figure 4, these distilled variants consistently reduced the gap to, and in some cases slightly exceeded, the original domain-specific teachers. This finding suggests that within-modality distillation can transfer domain-specific knowledge even without increasing data diversity.

In the third setting, we replaced the domain-specific teachers with generalist models trained on the broader PMC-OA database, including BiomedCLIP, BMCA, GenMedClip and UniMedCLIP, while still restricting student training to target-modality data. As the number of generalist teachers increased, student performance improved further and exceeded both the first-stage CLIP model and the single-modality distillation baselines on both X-ray and Pathology benchmarks. These observations suggest that generalist teachers may provide broader semantic structure learned from heterogeneous data, which remains beneficial even when student training data are modality-restricted.

In the fourth setting, we trained the student on the full PMC-OA database with joint distillation from all nine teachers, including both modality-specific and generalist models. This full MMKD-CLIP configuration achieved the highest AUC among the evaluated settings on both benchmarks and outperformed all single-modality and single-source distillation variants. Relative to the third setting, the additional gains are consistent with a combined contribution from mixed-modality training data and a hybrid teacher ensemble.

Overall, these results suggest that multimodality integration during both data pretraining and multi-teacher distillation contributes to the development of a biomedical foundation model with broader transfer ability. When training is restricted to a single modality, learned representations remain more modality-specific and may capture less cross-domain structure. By contrast, training on the full PMC-OA database with a hybrid teacher ensemble is associated with stronger performance on both

benchmarks. This improvement may reflect not only increased data scale but also complementary alignment across modalities: radiological images such as X-ray and CT emphasize macroscopic anatomy and spatial organization, whereas pathology images capture microscopic texture and cellular morphology. Joint distillation may therefore help the model align global and local visual information across domains.

Clinically, this multimodal integration is relevant. Real-world diagnostic workflows often require correlating findings across heterogeneous imaging sources (e.g., CT with histopathology, or ultrasound after MRI). A foundation model exposed to such cross-modal relations may develop modality-invariant yet clinically discriminative features, which may improve its adaptability to unseen datasets and composite diagnostic tasks.

Comparison of modality-specific and generalist ensemble strategies in zero-shot and linear-probing tasks

Following the ensemble formulation, we systematically compared our multi-teacher distilled model (MMKD-CLIP) against direct logit-level ensembles on both zero-shot classification and 10% linear-probing tasks (Supplementary Figures 5–6). The distilled model consistently shows performance that is comparable to or higher than ensemble combinations, along with greater stability than individual ensemble configurations.

On MRI, our method shows the highest AUC among the evaluated configurations, outperforming all single-teacher models and multi-teacher ensembles, including the highest-performing baseline formed by PMCCLIP and GenMedClip. A similar trend is observed for Ultrasound, where its accuracy matches the top-performing BiomedCLIP and the BMCA+BiomedCLIP ensemble, while exhibiting more stable performance across different teacher subsets. For X-ray, our model performs higher than domain-specific ensembles, indicating that distilling knowledge from heterogeneous modalities is associated with more balanced and transferable representations than directly merging specialized models. In Pathology, where particularly strong domain-specific models exist, ensembles such as PLIP+QuiltNet achieve the highest accuracy; our distilled model shows slightly lower yet comparable performance within this domain while maintaining consistent results across other modalities—a capability that specialized ensembles do not consistently provide.

Ensemble performance proved unstable and sensitive to teacher composition. For instance, on Ultrasound, different pairings of generalist CLIPs produced divergent outcomes: some combinations yielded moderate gains, while others resulted in substantial performance degradation. This sensitivity indicates that ensemble success depends strongly on the choice of teacher subsets and fusion weights, incurring trial-and-error costs for real-world deployment. In contrast, our unified model integrates complementary knowledge during training, providing stable performance without requiring post-hoc ensemble tuning.

Together, these results suggest that direct logit-level ensembling of multiple CLIP models—whether domain-specific or generalist—does not consistently provide stronger or more practical performance than the multi-teacher distillation approach. While naive ensembling can occasionally exploit complementary cues within a single modality, its benefits are inconsistent and limited in scope. By consolidating diverse visual-textual expertise, the final model shows improved robustness, cross-modality generalization, and scalability, making it suitable for deployment in biomedical imaging applications.

Impact of teacher-selection thresholds and soft weighting on distillation quality

To further examine the effect of the >90% confidence threshold used in our trustworthy teacher selection pipeline, and to assess whether softer weighting schemes might preserve harder examples while mitigating noise, we conducted a series of controlled experiments. The motivation behind this analysis is to clarify whether lowering the confidence cutoff enriches the training signal with informative hard cases.

The distinction between “hard” and “noisy” cases is central to this analysis. In biomedical CLIP models, zero-shot predictions with low confidence are not typically indicative of difficult but meaningful samples. Instead, they often arise from cross-modality leakage, where a teacher produces a superficially correct prediction for a mismatched domain. For example, a pathology model assigning a correct class label to an ultrasound sample by chance, but with a confidence that falls below reliable levels. Lowering the threshold therefore increases the number of such mismatches included in the distillation quadruplet, introducing noise into the training signal. To quantify this effect, we varied the confidence threshold from 90% to 60%, and additionally implemented a soft-weighting scheme in which all teacher outputs are retained and weighted proportionally to their softmax confidence. The results, summarized in Supplementary Figure 8, show that reducing the threshold to 60% consistently degrades zero-shot classification performance across all nine imaging modalities. This behavior suggests that low-confidence “correct” predictions are unlikely to correspond to beneficial hard cases but instead reflect unreliable cross-domain signals that reduce feature quality when incorporated into the student model. Soft weighting, although more flexible than thresholding, performs even worse. By weighting all teacher predictions, including unreliable ones, according to softmax confidence, this scheme does not fully filter out cross-domain mismatches. As a consequence, noise may accumulate throughout the dataset and propagate through the distillation loss, yielding performance that is generally below both threshold-based strategies. In contrast, the 90% threshold reduces such mismatches while retaining informative samples, leading to stable performance across modalities.

Visualizing and interpreting cross-modality alignment in distilled embeddings

To examine how cross-modality distillation contributes to the effectiveness of MMKD-CLIP, we visualized the joint image-text embedding spaces of MMKD-CLIP and nine biomedical CLIP baselines using t-SNE across diverse imaging modalities (Supplementary Figure 9). MMKD-CLIP consistently produces compact visual-textual clusters with closer alignment, suggesting that heterogeneous clinical cues are represented within a shared semantic manifold. In contrast, several baseline models exhibit fragmented or modality-separated structures in which image and text embeddings show increased separation, as highlighted in red. These geometric discrepancies suggest that single-modality or domain-limited CLIP models tend to form biased or less well-calibrated manifolds, whereas MMKD-CLIP maintains alignment across modalities with varying characteristics. The improved cross-modal embedding consistency provides a potential explanation for improved robustness and generalization across imaging specialties.

Discussion

MMKD-CLIP addresses an important limitation of existing biomedical vision-language models, namely their limited ability to generalize across heterogeneous and fragmented medical data, by introducing a multi-teacher distillation framework that unifies specialist knowledge into a single generalist representation. The proposed two-stage pipeline first performs CLIP-style pretraining on 2.9 million biomedical image-text pairs, followed by distillation from nine domain-specific and generalist biomedical CLIP teachers using a 16.5 million-pair offline multimodal corpus. This design enables aggregation of complementary expertise into a shared embedding space, allowing the student model to acquire modality-agnostic semantic understanding while maintaining domain-specific characteristics. Extensive evaluation across 58 datasets, spanning six task categories and nine imaging modalities, demonstrates stable performance of this approach. In particular, MMKD-CLIP consistently shows higher performance than individual teacher models in zero-shot classification, surpassing individual teachers in seven of nine modalities. These results help address the tendency of prior biomedical CLIP models to overfit narrowly curated domains and show that integrating knowledge from multiple expert teachers yields a single model capable of handling diverse clinical imaging scenarios, including MRI, CT, X-ray, endoscopy, pathology, OCT, and fundus imaging, without additional fine-tuning. Collectively, these findings support the use of MMKD-CLIP as a foundation model framework that can incorporate future advances in biomedical vision-language modeling.

An advantage of the proposed multi-teacher distillation strategy is its ability to deliver improved downstream performance under limited supervision. In the linear probing experiments (Figure 3a), particularly under low-data regimes (1% and 10% of training samples), MMKD-CLIP consistently maintains higher AUC values than baseline models (e.g., 5.46% in MRI at 1% data). These results indicate that the multi-teacher distillation strategy allows MMKD-CLIP to leverage complementary strengths from each teacher, such as PUBMEDCLIP's fine-grained subfigure-subcaption alignments, and QuiltNet and PLIP's histopathology expertise, which produces a student requiring less task-specific supervision. In practical terms, this means that clinics or resource-constrained settings could deploy MMKD-CLIP for classification tasks with limited labeling effort, an important advantage in low- and middle-income regions where medical annotations are costly and scarce.

Beyond downstream classification, MMKD-CLIP also exhibits improved cross-modal alignment between visual and textual representations. Through cross-modal retrieval experiments (Figure 3b), the model shows consistent improvements across both noisy and curated datasets. On the MedTrinity-25M subset, MMKD-CLIP achieves a text-to-image Recall@50 of 15.16%, representing a 1.49% improvement over the highest-performing baseline, and an image-to-text Recall@50 of 16.63%, a 1.57% gain. Meanwhile, on the cleaner BookSet, MMKD-CLIP reaches a Recall@50 of 52.95%, exceeding UniMedCLIP by 0.59%. These dual gains suggest the utility of MMKD-CLIP as a backbone architecture for both chart searches and real-time case-query systems, an important capability for a generalist biomedical model operating across varied clinical repositories and textual conventions.

The benefits of unified multimodal representation learning are further reflected in biomedical visual question answering performance. Across both SLAKE and VQA-RAD benchmarks (Figure 4), MMKD-CLIP shows the highest overall accuracy among evaluated models, indicating consistent recognition of anatomical structures and pathological findings, including spatial relationships and clinically relevant descriptors. Qualitative analyses suggest that the model maintains consistent reasoning even under ambiguous phrasing or partial visual occlusion, suggesting that the distilled joint embedding space captures both local visual cues and broader clinical context. This integration is consistent with patterns observed in clinical reasoning, where diagnostic decisions often require synthesizing information from imaging, pathology, and narrative reports.

Importantly, MMKD-CLIP extends beyond classification and retrieval to prognostic modeling. Survival prediction on TCGA (Figure 5) further extends MMKD-CLIP's utility into prognostic modeling. By fusing pathology patches with clinical reports, MMKD-CLIP shows higher C-Index values (e.g., 0.6873 versus PLIP's 0.6675). This improvement suggests that a unified vision-language representation can capture complementary prognostic signals (morphologic patterns in histology and clinical notes) that neither modality can independently convey. In effect, MMKD-CLIP learns a single model capable of

mapping multiple data sources into a risk-score prediction, supporting the development of comprehensive data-driven clinical decision support.

By distilling complementary strengths from multiple pathology teachers, MMKD-CLIP establishes a unified backbone that fine-tunes effectively for cancer diagnosis. In supervised cancer diagnosis tasks (Figure 6a), MMKD-CLIP's fine-tuned AUCs, ranging from 87.42% on BRACS-3 to 97.28% on TCGA-NSCLC, suggest that a distilled model can match or exceed specialist teacher performance when provided with a modest amount of labeled data. MMKD-CLIP's low variance across folds suggests that its pretrained weights are less sensitive to sampling noise, which is a recurring challenge in medical imaging where dataset sizes vary widely. By outperforming domain-specific models (e.g., QuiltNet for histopathology, PLIP for pathology) and generalist variants (e.g., GenMedClip, BMCA), MMKD-CLIP supports the objective of this study: providing a unified foundation that can be consistently fine-tuned for diverse downstream tasks without requiring multiple separate models.

Despite these strengths, MMKD-CLIP remains subject to several important considerations. As with any distillation-based framework, the student model may implicitly inherit biases present in the teacher models, including modality imbalances, institution-specific imaging characteristics, or annotation inconsistencies. While aggregating multiple teachers helps mitigate reliance on any single source, it does not fully eliminate domain-specific biases embedded in their training data. Addressing these issues will require future work on bias-aware distillation strategies, broader inclusion of underrepresented modalities, and prospective evaluation in clinical environments.

In summary, MMKD-CLIP is a generalist biomedical foundation model that unifies diverse imaging modalities and narrative contexts. Through multi-teacher distillation, we show that integrating expert knowledge from multiple domain-specific CLIP models yields a student with higher accuracy and improved robustness compared with individual teachers. Extensive evaluations across 58 datasets and six task categories support that MMKD-CLIP provides strong performance across biomedical vision-language tasks, offering a platform for future clinical and research applications.

Methods

Ethics statement

This study used only publicly available, de-identified benchmark datasets and did not involve direct human participant recruitment or access to identifiable information. Therefore, no additional institutional ethics approval was required for this study. Informed consent was not required for this secondary analysis because all data used in this study were publicly available and de-identified.

Pretraining data curation

For pretraining, we directly utilize the PMC-OA dataset curated by the BMCA¹⁹ project. Specifically, we adopt the “Concept-Filtering” subset, which comprises approximately 6 million image-text pairs. To further refine the dataset, we apply an additional filtering step that retains only those pairs annotated with an “image_primary_label” of “Microscopy” or “Clinical Imaging.” This results in a total of 2,911,190 pairs, including 1,416,257 single-panel images and 1,494,933 multi-panel images. In Supplementary Table 2, we summarize the distribution of image-text pairs across 26 imaging modalities represented in the filtered subset.

Model design and pretraining

The pre-training and distillation of MMKD-CLIP is inspired by CLIP-KD²¹ and consists of two main steps. The first step utilizes approximately 2.9 million image-text pairs collected from the PMC-OA database and applies contrastive learning to align the semantics of the two modalities. In the second step, for each image-text pair, we introduce four additional unpaired text samples and perform zero-shot classification using nine biomedical CLIP models. A CLIP model is deemed a trustworthy teacher if it achieves over 90% confidence on the ground-truth class. This pipeline yields a distillation-ready quadruplet comprising the original image, original text, and the corresponding visual and textual features extracted by the teacher model. Ultimately, this pipeline yields a total of 16,535,905 quadruplet samples. As illustrated in Figure 1c, knowledge distillation is then performed using both FD loss and ICL loss. The network backbone includes a visual encoder based on the MetaCLIP ViT-B/16 model³¹, and a text encoder based on the BioMed-BERT text encoder³². Model configurations are specified in Supplementary Table 1.

Multi-teacher dimension alignment

BMCA¹⁹ and PMCCLIP¹⁸ produce outputs of 768 dimensions, whereas other CLIP models output 512 dimensions, which requires dimensional alignment for distillation. Therefore, as illustrated in Supplementary Figure 1b, we constructed a dual-stream autoencoder for dimensional alignment. Specifically, we prepared a projection encoder for each CLIP to map their embeddings to the same dimension, with a target dimensionality of 512. The projected features of all CLIP models are then fed into a shared dual-stream autoencoder to learn a joint feature space. Finally, we designed a projection decoder for each CLIP to

reconstruct the features. We do not implement contrastive learning loss in the latent space, as this may alter the distribution of the original feature space of each CLIP.

The formalized description of the MMKD-CLIP pipeline

In the pretraining phase, given a multimodal dataset containing N image–text pairs, denoted as

$$\mathcal{D} = \{(I_1, T_1), \dots, (I_i, T_i), \dots, (I_N, T_N)\},$$

where I_i represents the i -th image and T_i represents the i -th caption, our goal is to develop a biomedical VLM using a visual encoder $f(\cdot)$ and a text encoder $g(\cdot)$ to perform image–text alignment, such that paired image–text samples are pulled closer together and unpaired samples are pushed farther apart in the embedding space. Specifically, given an image–text pair (I_i, T_i) , the visual encoder $f(\cdot)$ and text encoder $g(\cdot)$ encode the pair as follows:

$$\mathbf{v}_i = f(I_i), \quad \mathbf{v}_i \in \mathbb{R}^d \quad (3)$$

$$\mathbf{t}_i = g(T_i), \quad \mathbf{t}_i \in \mathbb{R}^d \quad (4)$$

Here, $\mathbf{v}_i$ represents the global feature vector of the image, $\mathbf{t}_i$ represents the global feature vector of the caption, and d denotes the dimension of the embedding space. All features are post-processed using ℓ_2 normalization.

CLIP-style pretraining employs a contrastive learning framework inspired by InfoNCE³³ to align visual and textual representations. Specifically, given an image embedding vector $\mathbf{v}_i$ and its corresponding text embedding $\mathbf{t}_i$, the objective is to maximize their similarity while minimizing similarity to mismatched pairs within a mini-batch. The image-to-text contrastive loss, where the image serves as the anchor, is defined as:

$$\mathcal{L}_{I \rightarrow T} = - \sum_{i=1}^{|\mathcal{B}|} \log \frac{\exp(\mathbf{v}_i \cdot \mathbf{t}_i / \tau)}{\sum_{b=1}^{|\mathcal{B}|} \exp(\mathbf{v}_i \cdot \mathbf{t}_b / \tau)} \quad (5)$$

Conversely, the model also optimizes a symmetric contrastive loss in the reverse direction, with the text embedding $\mathbf{t}_i$ as the anchor. The corresponding text-to-image loss is formulated as:

$$\mathcal{L}_{T \rightarrow I} = - \sum_{i=1}^{|\mathcal{B}|} \log \frac{\exp(\mathbf{t}_i \cdot \mathbf{v}_i / \tau)}{\sum_{b=1}^{|\mathcal{B}|} \exp(\mathbf{t}_i \cdot \mathbf{v}_b / \tau)} \quad (6)$$

Here, the operator “ $\cdot$ ” denotes the dot product between embedding vectors, which is equivalent to cosine similarity because the embeddings are ℓ_2 -normalized, and τ is a learnable temperature parameter that scales the logits. The summations are computed over all samples within a mini-batch $\mathcal{B}$. The final training objective of CLIP integrates both directions by averaging the image-to-text and text-to-image losses:

$$\mathcal{L}_{\text{CLIP}} = \frac{1}{2} (\mathcal{L}_{I \rightarrow T} + \mathcal{L}_{T \rightarrow I}) \quad (7)$$

In the knowledge distillation phase, to narrow the performance discrepancy between teacher and student models, a straightforward yet effective method is to align their intermediate feature representations, referred to as feature distillation (FD). The core idea is that if the student’s features approximate those of the teacher, the performance gap may be reduced. Therefore, we encourage the student to replicate both the image and text embeddings of the teacher by minimizing the mean squared error (MSE) loss, defined as:

$$\mathcal{L}_{\text{FD}} = \frac{1}{|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} \left(\|\mathbf{v}_i^T - \mathbf{v}_i^S\|_2^2 + \|\mathbf{t}_i^T - \mathbf{t}_i^S\|_2^2 \right) \quad (8)$$

Here, $\mathbf{v}_i^T$ and $\mathbf{t}_i^T$ represent the teacher’s visual and textual feature vectors for the i -th instance, whereas $\mathbf{v}_i^S$ and $\mathbf{t}_i^S$ denote those of the student. The loss aggregates the squared ℓ_2 distances across all samples in the mini-batch $\mathcal{B}$.

To enhance knowledge transfer between the teacher and student models, we introduce an approach termed *Interactive Contrastive Learning* (ICL), which establishes contrastive relationships between cross-model embeddings. Specifically, the student encoder outputs are contrasted against teacher embeddings rather than against embeddings from the same network. When using the student’s visual representation $\mathbf{v}_i^S$ as the query and teacher text embeddings $\{\mathbf{t}_b^T\}_{b=1}^{|\mathcal{B}|}$ as targets, the image-to-text contrastive objective under the ICL framework is defined as:

$$\mathcal{L}_{\text{ICL}, I \rightarrow T} = - \sum_{i=1}^{|\mathcal{B}|} \log \frac{\exp(\mathbf{v}_i^S \cdot \mathbf{t}_i^T / \tau)}{\sum_{b=1}^{|\mathcal{B}|} \exp(\mathbf{v}_i^S \cdot \mathbf{t}_b^T / \tau)} \quad (9)$$

In a reciprocal manner, for student-generated text features $\mathbf{t}_i^S$, the corresponding teacher visual embeddings $\{\mathbf{v}_b^T\}_{b=1}^{|\mathcal{B}|}$ are used to compute the text-to-image contrastive loss:

$$\mathcal{L}_{\text{ICL}, T \rightarrow I} = - \sum_{i=1}^{|\mathcal{B}|} \log \frac{\exp(\mathbf{t}_i^S \cdot \mathbf{v}_i^T / \tau)}{\sum_{b=1}^{|\mathcal{B}|} \exp(\mathbf{t}_i^S \cdot \mathbf{v}_b^T / \tau)} \quad (10)$$

The overall ICL objective integrates both directional losses by averaging:

$$\mathcal{L}_{\text{ICL}} = \frac{1}{2} (\mathcal{L}_{\text{ICL}, I \rightarrow T} + \mathcal{L}_{\text{ICL}, T \rightarrow I}) \quad (11)$$

Optimizing the ICL loss can be interpreted as increasing the lower bound on mutual information between student and teacher representations. CLIP-KD²¹ has proved this theory. This mutual information perspective suggests that as the student learns to align with the teacher’s embeddings, particularly when contrasting against teacher outputs, it gains reduced uncertainty and better representation quality. This dynamic encourages deeper knowledge sharing from teacher to student networks.

To unify the objectives in the distillation phase, we adopt a weighted combination of the aforementioned losses. Specifically, we retain the original CLIP loss with a smaller weight to maintain modality alignment, and integrate both the feature distillation loss and the interactive contrastive loss to fully facilitate knowledge transfer. The overall distillation objective is expressed as:

$$\mathcal{L}_{\text{KD}} = \alpha_1 \cdot \mathcal{L}_{\text{CLIP}} + \alpha_2 \cdot \mathcal{L}_{\text{FD}} + \alpha_3 \cdot \mathcal{L}_{\text{ICL}} \quad (12)$$

Here, α_1 , α_2 , and α_3 are weighting coefficients, set to 0.1, 50, and 1, respectively.

Zero-shot and linear probing image classification

We evaluate the zero-shot and linear probing image classification abilities of MMKD-CLIP alongside previously reported medical CLIP models across 38 datasets covering nine imaging modalities. In the zero-shot scenario, ground-truth labels are not used during inference. We follow the zero-shot protocol described in CLIP², in which labels are combined with prompt words to form sentences, and similarity is computed between the image and the corresponding positive and negative sentence samples. For both positive and negative samples, we generate multiple sentences for each label using various prompts and aggregate these by averaging their embeddings to accommodate descriptive variations. The class corresponding to the highest image–text similarity is then designated as the final classification result. For linear probing, we utilize the frozen MMKD-CLIP to extract embedded features for each image and employ a lightweight linear classifier for classification. The classifier is optimized using the training set of downstream datasets and then evaluated on their test sets. We keep the test set unchanged while varying the proportion of the training set used during optimization to evaluate generalization under limited training data. We use the AUC score for evaluation to reduce dependence on threshold selection.

Cross-modal retrieval

To evaluate whether MMKD-CLIP aligns image and text concepts in the latent space, we perform image-to-text and text-to-image retrieval. In this task, we freeze MMKD-CLIP to compute embedding features for all images and texts, and then use cosine similarity to determine the similarity between each possible pair. For text-to-image retrieval, we retrieve the top-K images closest to a given text query in the aligned latent space. Similarly, image-to-text retrieval follows an analogous procedure. We utilize the Recall@K metric to evaluate retrieval performance. This metric calculates the proportion of test set candidates where the correct match is found within the top-K retrieved samples. We use K values of {1, 10, 50}.

Visual question answering

Visual question answering (VQA) is a common task for evaluating human–computer interaction. Given an image and a natural language question, the model generates an answer to the question. We use the framework provided by PUBMEDCLIP³⁴ to facilitate experiments on the VQA task, which treats VQA as a classification task. We replace the image and text encoders in this framework with frozen MMKD-CLIP or various other biomedical CLIP models, while keeping the rest unchanged. All models were fine-tuned using the QCR³⁴ (question answering via conditional reasoning) framework. Finally, we use closed accuracy, open accuracy, and overall accuracy to evaluate model performance.

Supervised cancer diagnosis

We use frozen MMKD-CLIP to extract visual features of pathology slides offline, which are input into the ABMIL model³⁵ for multiple instance learning (MIL). As shown in Figure 1d, the output of the model is the probability distribution of cancer categories. We use AUC as the evaluation metric to reduce the impact of threshold selection.

Survival prediction

We evaluated MMKD-CLIP on a survival prediction task using a combination of pathology images and clinical reports. To this end, we used the slice dataset (0.5 $\mu\text{m}/\text{pixel}$) provided by the TCGA-UT²⁷, which involves 32 types of cancer. In addition, following the practice of MUSK³⁶, we matched the TCGA-UT dataset with TCGA-reports³⁷ and eliminated cancer types with limited patient counts, ultimately selecting 12 cancer types: BRCA, LUAD, LUSC, KIRC, UCEC, HNSC, BLCA, STAD, SARC, CESC, PAAD, and MESO. We used ABMIL³⁵ to aggregate image features and trained prognostic models for each cancer type based on multiple instance learning (MIL). Since this task predicts the probability of survival time falling into predefined intervals, it can be regarded as a classification task. We evaluated the model through five-fold cross-validation and used the concordance index (C-Index) as the evaluation metric.

Ablation of loss components

We perform an ablation study on the zero-shot image classification task across four major imaging modalities: X-ray, pathology, CT, and MRI. Furthermore, we compare the overall performance of each component on all 38 datasets (“Overall Dataset”). As shown in Figure 6b, we compare four settings: (i) pretraining only without distillation, (ii) full model without the FD loss, (iii) full model without the ICL loss, and (iv) the full MMKD-CLIP with all components.

Modality ensemble for evaluating multi-teacher distillation

To determine whether multi-teacher distillation provides advantages beyond simple model ensembling, we established a formal ensemble framework applicable to both zero-shot classification and linear probing with 10% labeled data (Supplementary Figures 5–6). Our evaluation employs two complementary ensemble strategies designed for modality-aware comparisons. For modalities with dedicated CLIP models, specifically X-ray and Pathology, we constructed domain-specific ensembles. In X-ray, ensembles combined MedCLIP and PUBMEDCLIP, optionally including UniMedCLIP; for Pathology, ensembles combined PLIP and QuiltNet, also optionally with UniMedCLIP. For modalities lacking dedicated single-modality CLIPs (MRI and Ultrasound), we formed generalist ensembles from four broadly trained biomedical CLIP models: BMCA, PMCCCLIP, BiomedCLIP, and GenMedClip. This strategy approximates the integrated cross-domain knowledge that multi-teacher distillation is designed to capture.

Formally, for an input image x and C textual class descriptions $\{t_1, \dots, t_C\}$, each teacher model m produces image and text features $f_{\text{img}}^m(x)$ and $f_{\text{text}}^m(t_c)$. These features are ℓ_2 -normalized:

$$\tilde{\mathbf{v}}^m(x) = \frac{f_{\text{img}}^m(x)}{\|f_{\text{img}}^m(x)\|_2}, \quad \tilde{\mathbf{u}}_c^m = \frac{f_{\text{text}}^m(t_c)}{\|f_{\text{text}}^m(t_c)\|_2}. \quad (13)$$

The cosine similarity between them, scaled by the teacher temperature τ_m , defines the logit for class c :

$$z_c^m(x) = \frac{\langle \tilde{\mathbf{v}}^m(x), \tilde{\mathbf{u}}_c^m \rangle}{\tau_m} = \frac{\tilde{\mathbf{v}}^m(x)^\top \tilde{\mathbf{u}}_c^m}{\tau_m}. \quad (14)$$

For an ensemble of M teachers with non-negative weights $\{w_m\}_{m=1}^M$ summing to one, we aggregate logits as:

$$z_c^{\text{ens}}(x) = \sum_{m=1}^M w_m z_c^m(x), \quad (15)$$

and apply a unified softmax to obtain the ensemble probability distribution:

$$p_{\text{ens}}(c | x) = \frac{\exp(z_c^{\text{ens}}(x))}{\sum_{j=1}^C \exp(z_j^{\text{ens}}(x))}, \quad \hat{y}_{\text{ens}}(x) = \arg \max_c p_{\text{ens}}(c | x). \quad (16)$$

The same formulation applies to the linear-probing setting, where each teacher’s projection head produces logits $z_c^m(x)$ that are averaged prior to evaluation. This formulation provides a principled and consistent framework for comparing direct logit-level ensembling with multi-teacher distillation under modality-aware conditions.

Biomedical CLIP models for comparison

MMKD-CLIP is distilled from nine previously reported biomedical CLIP models. Therefore, we selected these nine models for comparison, including BMCA¹⁹, GenMedClip³⁸, MedCLIP³⁹, BiomedCLIP⁴⁰, UniMedCLIP⁴¹, PLIP¹⁰, PUBMEDCLIP³⁴, QuiltNet⁴², and PMCCLIP¹⁸.

- BMCA is trained on the BIOMEDICA dataset, which includes over 24M image-text pairs from over 6M open-access scientific articles.
- GenMedClip is a vision-language model built on the CLIP architecture, pretrained on 4.7M medical image-text pairs spanning 12 imaging modalities. It is trained using the MedicalNarratives dataset, which captures synchronized instructor speech and mouse-tracking data from pedagogical videos, providing semantic and dense annotations (e.g., traces and bounding boxes) to support learning across diverse medical imaging tasks.
- MedCLIP addresses the scarcity of large-scale paired datasets and the prevalence of false negatives in traditional contrastive learning. By decoupling images and text, MedCLIP combines and expands training pairs using a semantic matching loss function based on medical knowledge. Even with only 10% of the training data, MedCLIP shows improved performance compared with previously reported models.
- BiomedCLIP is a multimodal foundation model pretrained on PMC-15M, a dataset of 15 million image-text pairs sourced from 4.4 million scientific articles.
- UniMedCLIP is a unified vision-language model trained on UniMed, an open-source dataset of 5.3 million image-text pairs spanning six imaging modalities (X-ray, CT, MRI, Ultrasound, Pathology, Fundus).
- PLIP is a multimodal vision-language model trained on OpenPath, a dataset of 208,414 de-identified pathology images with natural language descriptions curated from public platforms such as medical Twitter.
- PUBMEDCLIP is a domain-adapted version of CLIP fine-tuned on PubMed articles for Medical Visual Question Answering (MedVQA). By leveraging medical image-text pairs, PUBMEDCLIP shows improved performance compared with prior methods such as MAML, achieving up to 3% higher accuracy on MedVQA benchmarks and suggesting the potential of contrastive pretraining in biomedical tasks.
- QuiltNet is trained on QUILT-1M, a vision-language dataset in histopathology built from 1,087 hours of expert YouTube videos and diverse web sources. By fine-tuning a pre-trained CLIP model on this dataset, QuiltNet enables improved zero-shot classification and cross-modal retrieval across 13 datasets spanning 8 histopathological subtypes.
- PMCCLIP is trained on PMC-OA, a 1.6M biomedical image-caption dataset curated from the PubMed OA database. Leveraging fine-grained subfigure-subcaption alignments, PMCCLIP achieves strong performance across retrieval, classification, and VQA tasks in the biomedical domain.

Benchmark datasets

We evaluated the MMKD-CLIP model across multiple medical AI tasks using 58 publicly available benchmark datasets spanning zero-shot classification, linear probing image classification, cross-modal retrieval, visual question answering, and outcome prediction. For zero-shot and linear probing image classification, we used 38 different medical imaging datasets across 9 modalities:

- X-Ray datasets include CovidCXR4⁴³, which comprises 84,818 chest X-ray images from 45,342 patients, labeled for binary classification to detect COVID-19 infection versus normal findings. DDSM⁴⁴ contains 71,249 pre-processed breast X-ray images, combining data from the DDSM and CBIS-DDSM collections. These images are labeled for binary classification to detect breast cancer, distinguishing between positive findings (malignant/suspicious lesions) and negative

findings (normal breast tissue). NLMTB⁴⁵ includes 4,200 chest X-ray radiographs from the U.S. National Library of Medicine, labeled for binary classification to detect tuberculosis versus normal chest findings. SIIMACR⁴⁶ is a chest X-ray dataset with 12,047 DICOM images labeled for binary classification and segmentation to detect pneumothorax, with pixel-level annotations for positive cases. RSNA Pneumonia⁴⁷ comprises 30,227 frontal-view chest radiographs labeled for binary classification to detect pneumonia, derived from the RSNA Pneumonia Detection Challenge 2018.

- CT datasets include BrainTumorCT⁴⁸, which comprises 4,618 high-resolution brain CT scans labeled for binary classification to distinguish between healthy brain tissue and brain tumors. CT Axial, CT Coronal, CT Sagittal⁴⁹ are three subsets from the MedIMeta collection, comprising 1,645 standardized abdominal CT image slices in axial, coronal, and sagittal planes. These images are labeled for 11-class organ classification, identifying anatomical structures such as the heart, lungs, liver, kidneys, spleen, pancreas, bladder, and femoral heads. CovidCT3A⁵⁰ is a large-scale dataset with 425,024 chest CT slices from 5,312 patients, labeled for 3-class classification to distinguish between normal lungs, pneumonia, and COVID-19 infection. OrganaMNIST⁵¹ includes 58,830 abdominal CT images pre-processed to 28×28 pixels from the MedMNIST collection, labeled for 11-class classification to identify different organs, including the bladder, femurs, heart, kidneys, liver, lungs, pancreas, and spleen.
- MRI datasets include BrainTumorMRI⁴⁸, which comprises 5,000 high-resolution brain MRI scans from multiple patients. This dataset is labeled for binary classification to distinguish between healthy brain tissue and the presence of brain tumors, leveraging MRI's superior soft tissue contrast for detailed analysis. BrainTumorMRI2⁵² contains 7,023 brain MRI images labeled for 4-class classification, distinguishing between glioma, meningioma, pituitary tumors, and healthy brain tissue. ACL MRI⁵³ includes 1,021 knee MRI examinations for binary classification of ACL (anterior cruciate ligament) tears.
- Ultrasound datasets include BUSBRA⁵⁴, which comprises 1,875 breast ultrasound images from 1,064 patients with biopsy-proven cases. The dataset is labeled for binary classification to distinguish between malignant and benign breast lesions. Breast US⁵⁵ contains 780 images from 600 female patients, also labeled for binary classification between benign and malignant breast lesions. BreastMNIST⁵¹ includes 780 breast ultrasound images pre-processed to 28×28 pixels from the MedMNIST collection. It is labeled for binary classification to differentiate between malignant and normal/benign breast conditions.
- Fundus datasets include DRD⁵⁶, which comprises 35,125 retinal photographs labeled for 5-class diabetic retinopathy severity grading, ranging from no diabetic retinopathy to proliferative diabetic retinopathy. FundusJSIEC⁵⁷ contains 1,000 retinal images from the Joint Shantou International Eye Centre, labeled for 39-class classification. This dataset covers a wide range of retinal diseases and conditions, including diabetic retinopathy, glaucoma, macular disorders, and vascular occlusions. FIVE Retina⁵⁸ includes 800 retinal photographs labeled for 4-class classification to detect age-related macular degeneration, diabetic retinopathy, glaucoma, and normal conditions, derived from the FIVES dataset.
- OCT datasets include OCTMNIST⁵¹, which comprises 109,309 optical coherence tomography images from the MedMNIST v2 collection. These images are pre-processed to a standardized 28×28 pixel grayscale format, facilitating lightweight classification tasks. The dataset is labeled for 4-class classification to identify choroidal neovascularization (CNV), diabetic macular edema (DME), drusen, and normal retinal conditions. RetinalOCT⁵⁹ includes 24,000 high-quality OCT images, labeled for 8-class classification. This dataset is used to detect a range of retinal conditions, including age-related macular degeneration (AMD), CNV, central serous retinopathy (CSR), DME, diabetic retinopathy (DR), drusen, macular hole (MH), and normal retinal conditions.
- Pathology datasets include BACH⁶⁰ with 400 breast histopathology images for 4-class classification, distinguishing between normal tissue, benign lesions, in-situ carcinoma, and invasive carcinoma. LC25000 Colon⁶¹ comprises 10,000 colon tissue images for binary classification of adenocarcinoma and benign tissue. LC25000 Lung⁶¹ includes 15,000 lung tissue images for 3-class classification, identifying adenocarcinoma, benign tissue, and squamous cell carcinoma. NCT CRC⁶² offers 107,180 colorectal cancer tissue patches for 9-class classification, covering various tissue types. Osteo⁶³ contains 1,091 osteosarcoma tissue tiles for 3-class classification, distinguishing non-tumor, necrotic, and viable tumor regions. PCAM⁶⁴ provides 327,680 lymph node images for binary classification of metastatic breast cancer presence. Kather et al. (2016)⁶⁵ features 5,000 colorectal cancer histology images for 8-class classification of tissue types. Kather et al. (2018)⁶⁶ includes 1,800 images for 9-class classification, derived from the NCT-CRC-HE-100K dataset. Kather et al. val7k (2018)⁶⁶ is a validation set with 1,314 images for 9-class classification. Skin Cancer⁶⁷ comprises 6,107 skin histopathology tiles for 16-class classification, including 12 non-tumor and 4 tumor categories. Tang et al. (2019)⁶⁸

offers 491 brain tissue images for 4-class classification of amyloid pathologies. Wong et al. (2022)⁶⁹ provides 800 brain tissue images for 4-class classification of amyloid pathologies, based on multi-expert annotations.

- Endoscopy datasets include Kvasir⁷⁰ with 8,000 gastrointestinal images for 8-class classification, covering anatomical landmarks and pathological findings, and WCE⁷¹ containing 6,000 wireless capsule endoscopy images across four categories, aimed at detecting normal tissue, ulcerative colitis, polyps, and esophagitis conditions.
- Dermatology datasets comprise HAM10000⁷² with 10,015 skin lesion images for 7-class classification and PADUFES20⁷³ containing 2,298 smartphone-captured lesions across six categories.

For cross-modal retrieval, BookSet²⁴ provides 4,265 image-text pairs from academic textbooks, while MedTrinity-25M subset²³ contains 3,771 multimodal medical samples. Visual question answering evaluation used VQA-RAD²⁶ with 3,515 radiological question-answer pairs and SLAKE²⁵ containing 14,000 questions across 642 medical images (Only the English portion was utilized in this study). Supervised cancer diagnosis employed Camelyon16³⁰ with 4,809,940 patches from 399 whole-slide lymph node images, TCGA-NSCLC²⁹ containing 4,377,051 patches from 1,046 lung cancer slides, and BRACS²⁸ with 45,390 breast histopathology patches across seven lesion categories. Survival prediction was evaluated on TCGA-UT²⁷ containing 138,220 patches from 12 cancer types using five-fold cross-validation. Detailed descriptions of each dataset are provided in Supplementary Data 2.

Implementation details

We initialize image encoder from MetaCLIP Vit-B/16³¹, and initialize text encoder from BioMed-BERT³². The size of each input image is 224×224 pixels. During the pretraining phase, we use the ADAMW⁷⁴ optimizer with a learning rate of $5e-5$. Training is conducted on two A100 GPUs with a batch size of 512 for 20 epochs, with the first 2000 steps serving as a warm-up phase. In the distillation stage, we ensure that each original image and text are traversed for 20 epochs, and each image-text pair extracts a teacher feature pair from its corresponding trustworthy teacher group for distillation. The batch size at this stage is 384.

For linear probing image classification task, we used a single-layer fully connected neural network with the input dimension being the CLIP feature dimension and the output dimension being the number of categories in the downstream task dataset. We use the ADAMW optimizer with a learning rate of $5e-5$. Training is conducted on one A6000 ADA GPU with a batch size of 128 for 20 epochs. For VQA task, we kept the same settings as PUBMEDCLIP³⁴. The number of epoch is set to 150. For survival prediction task, we estimate the survival function using the Kaplan-Meier method, dividing event times into equal probability intervals to obtain time interval cut points that reflect event density. Then, we used ADAM as the optimizer with a learning rate of $1e-4$ and performed 50 epochs of training. When visualizing, we grouped individuals into high-risk and low-risk categories based on the median of the predicted risk score.

Metrics

We employ 7 widely-recognized evaluation metrics to assess MMKD-CLIP's performance across classification, retrieval, VQA, and survival prediction tasks: AUC, Accuracy, Recall, C-Index, $C\text{-Index}_{td}$, and IBS.

The AUC metric quantifies a classifier's capability to distinguish between positive and negative classes across various decision thresholds. It is particularly important in medical classification tasks for its threshold-independent nature and ability to summarize diagnostic performance.

Accuracy (ACC) reflects the proportion of correct predictions among all predictions made. It is especially relevant in visual question answering, where it captures the model's ability to select the correct answer from multiple choices. It is defined as:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (17)$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

Recall quantifies the model's ability to retrieve all relevant items from the total pool of ground-truth positives. In our retrieval setting, we report Recall@K, where $K \in \{1, 10, 50\}$, indicating the proportion of cases in which the correct match appears within the top-K retrieved results. This helps assess performance at varying tolerance levels for top-ranked predictions. Formally, recall is defined as:

$$Recall = \frac{TP}{TP + FN} \quad (18)$$

Higher Recall@1 suggests stronger performance at the top-ranked prediction, while Recall@10 and Recall@50 reflect broader retrieval robustness.

The concordance index (C-index) assesses how well a model predicts the relative order of event times. It is commonly used in survival analysis to evaluate the model's ranking capability. The C-index is computed as the proportion of all comparable subject pairs for which the predicted risk scores and the observed event ordering are concordant. Formally, it is defined as

$$\text{C-index} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \mathbb{I}(\hat{r}_i > \hat{r}_j), \quad (19)$$

where $\mathcal{P}$ is the set of comparable pairs such that subject i experienced the event before subject j , $\hat{r}_i$ and $\hat{r}_j$ are the predicted risk scores for subjects i and j , respectively, and $\mathbb{I}(\cdot)$ is the indicator function, which equals 1 when the predicted ordering agrees with the observed ordering and 0 otherwise. A C-index of 1 indicates perfect concordance, whereas 0.5 implies random prediction.

The time-dependent concordance index (C-index_{td}) generalizes the traditional C-index by evaluating the model's ability to rank event risks at multiple time horizons. Unlike the standard version, it incorporates time-specific risk estimates, making it suitable for dynamic prediction settings. A general form can be written as

$$\text{C-index}_{\text{td}} = \frac{\sum_{t \in \mathcal{T}} \sum_{(i,j) \in \mathcal{P}_t} \mathbb{I}(\hat{r}_i(t) > \hat{r}_j(t))}{\sum_{t \in \mathcal{T}} |\mathcal{P}_t|}, \quad (20)$$

where $\mathcal{P}_t$ denotes the set of comparable pairs at time t , and $\hat{r}_i(t)$ is the predicted risk score for individual i at time t .

The Integrated Brier Score (IBS) measures the accuracy of survival probability estimates over a continuous time range. It is defined as the time-integrated mean squared difference between the predicted survival probability and the observed survival status. Lower values indicate better performance. Formally, it is given by

$$\text{IBS} = \frac{1}{\tau} \int_0^\tau \mathbb{E} \left[(\hat{S}(t | \mathbf{x}) - \mathbb{I}(T_i > t))^2 \right] dt \quad (21)$$

where $\hat{S}(t | \mathbf{x})$ is the predicted survival probability at time t for feature vector $\mathbf{x}$, and $\mathbb{I}(T_i > t)$ is the indicator of whether subject i remains event-free beyond time t .

Statistics and Reproducibility

Statistical analyses were tailored to the structure of each evaluation task. For evaluations performed on a single test set, including cross-modal retrieval, visual question answering, and supervised classification, uncertainty was quantified using nonparametric bootstrap resampling with replacement. Specifically, 1,000 bootstrap resamples were drawn from the corresponding independent test units, and 95% confidence intervals (CIs) were computed from the empirical bootstrap distribution. The independent unit of analysis was defined according to the task: individual queries for retrieval, question-answer pairs for VQA, and test samples from the evaluation cohort for supervised classification. For modality-level zero-shot and ablation analyses that aggregate results across multiple datasets, each dataset was treated as one independent experimental unit. Within each dataset, performance was first re-estimated using 1,000 bootstrap resamples with replacement. For each bootstrap iteration $b \in \{1, \dots, 1000\}$, the resampled performance values were averaged across datasets within the same group or modality to obtain a group-level bootstrap replicate:

$$\bar{\theta}^{(b)} = \frac{1}{K} \sum_{k=1}^K \theta_k^{(b)}. \quad (22)$$

Here, K is the number of datasets in the group, and $\theta_k^{(b)}$ denotes the performance estimate from the b -th bootstrap resample of dataset k . The 95% confidence interval (CI) was defined by the 2.5th and 97.5th percentiles of the resulting 1,000 group-level bootstrap replicates $\{\bar{\theta}^{(b)}\}_{b=1}^{1000}$.

Statistical significance between two models was assessed using two-sided empirical paired bootstrap tests based on the same bootstrap resamples. Let $\theta_A^{(b)}$ and $\theta_B^{(b)}$ denote the bootstrap performance estimates of models A and B at iteration b , and define the paired difference as

$$\Delta^{(b)} = \theta_A^{(b)} - \theta_B^{(b)}. \quad (23)$$

The two-sided empirical bootstrap P value was calculated as

$$P = 2 \times \min \left(\frac{1}{B} \sum_{b=1}^B \mathbb{I}(\Delta^{(b)} \leq 0), \frac{1}{B} \sum_{b=1}^B \mathbb{I}(\Delta^{(b)} \geq 0) \right), \quad (24)$$

where $B = 1000$. This tests whether the paired bootstrap distribution of performance differences is consistently above or below zero. With 1,000 bootstrap replicates, the smallest non-zero empirical two-sided P value is approximately 0.002. Therefore, when the estimated P value is less than or equal to 0.002, we report it as $P = 0.002$ after rounding.

For experiments based on five-fold cross-validation, performance was summarized as the mean across folds, and variability was reported as the standard deviation across folds. These fold-wise results reflect variation across data splits rather than independent biological replicates. When overall comparisons were made across multiple cohorts, each cohort was treated as one independent experimental unit, and statistical significance was assessed across cohorts using a two-sided Wilcoxon signed-rank test. For time-to-event analyses, prognostic performance was evaluated using the concordance index (C-Index). Kaplan–Meier survival curves were used to visualize risk stratification, with patients divided into high-risk and low-risk groups according to the median predicted risk score within each cohort. Differences between survival curves were assessed using two-sided log-rank tests.

No statistical method was used to predetermine sample size. Sample sizes were determined by the size of the publicly available benchmark test sets used in each task. No data were excluded from the analyses unless exclusion criteria were explicitly defined by the original dataset providers. The experiments were not randomized because this study evaluated fixed retrospective benchmark datasets and predefined data splits. The investigators were not blinded to allocation during experiments and outcome assessment because all evaluations were computational and performed on predefined public datasets using fixed protocols.

All experiments were conducted using publicly available datasets with predefined or explicitly described evaluation splits, and all reported results were generated using the same analysis pipeline and fixed evaluation protocols. The code used for model training, evaluation, and statistical analysis is publicly available, enabling reproduction of the reported results.

Data Availability

The data used for pre-training is available from the Huggingface project of BIOMEDICA (<https://huggingface.co/BIOMEDICA>). The data used for distillation can be obtained from MMKD-CLIP⁷⁵ (<https://github.com/wangshansong1/MMKD-CLIP>). All datasets used in this study are publicly available through established biomedical repositories or institutional portals, and can be accessed at the following sources: BrainTumorCT/MRI (<https://www.kaggle.com/datasets/murtozalikhon/brain-tumor-multimodal-image-ct-and-mri>; <https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>); CovidCT3A (<https://www.kaggle.com/datasets/hgunraj/covidxct>); CT Axial/Coronal/Sagittal (MedIMeta) (<https://arxiv.org/abs/2404.16000>); OrganaMNIST / BreastMNIST / OCTMNIST (<https://medmnist.com/>); HAM10000 (<https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000>); PAD-UFES-20 (<https://smoggy-whimsey-274.notion.site/PAD-UFES-20-c1e0ff8f36814b94b205ee919e2e3ff8>); Kvasir (<https://datasets.simula.no/kvasir/>); WCE curated colon dataset (<https://www.kaggle.com/datasets/francismon/curated-colon-dataset-for-deep-learning>); DRD (<https://www.kaggle.com/competitions/diabetic-retinopathy-detection>); FundusJSIEC (<https://www.kaggle.com/datasets/linchundan/fundusimage1000>); FIVE Retina (<https://www.kaggle.com/datasets/mohamedbounechada/fives-a-fundus-image>); ACL MRI (MRNet) (<https://github.com/ahmedbesbes/mrnet>); RetinalOCT (<https://www.kaggle.com/datasets/obulisainaren/retinal-oct-c8>); BACH (<https://zenodo.org/records/3632035>); LC25000 (Colon and Lung) (https://github.com/tampapath/lung_colon_image_set); NCT-CRC-HE (<https://zenodo.org/records/1214456>); Kather pathology datasets (2016, 2018, val7k) (<https://pubmed.ncbi.nlm.nih.gov/27306927/>; <https://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.1002730>); Osteosarcoma tiles (<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0210706#sec001>); PCAM (PatchCamelyon) (<https://www.kaggle.com/datasets/andrewmvd/metastatic-tissue-classification-patchcamelyon>); Skin Cancer Heidelberg (<https://heidata.uni-heidelberg.de/dataset.xhtml?persistentId=doi:10.11588/data/7QCR8S>); Tang et al. 2019 (<https://zenodo.org/records/1470797>); Wong et al. 2022 (<https://osf.io/xh2jd/>); BUSBRA (<https://zenodo.org/records/8231412>); Breast Ultrasound (<https://pubmed.ncbi.nlm.nih.gov/31867417/>); CovidCXR4 (<https://www.kaggle.com/datasets/andyczhao/covidx-cxr2>); DDSM / CBIS-DDSM (<https://www.kaggle.com/datasets/skooch/ddsm-mammography>); NLMTB

(NLM Tuberculosis) (<https://pmc.ncbi.nlm.nih.gov/articles/PMC4256233/>); SIIM-ACR Pneumothorax (<https://www.kaggle.com/c/siim-acr-pneumothorax-segmentation>); RSNA Pneumonia Detection Challenge (<https://www.rsna.org/rsnai/ai-image-challenge/rsna-pneumonia-detection-challenge-2018>); BookSet (https://warwick.ac.uk/fac/cross_fac/tia/data/arch); MedTrinity-25M subset (<https://huggingface.co/datasets/UCSC-VLAA/MedTrinity-25M>); Camelyon16 (<https://camelyon16.grand-challenge.org/>); TCGA-NSCLC (<https://www.cancer.gov/ccg/research/genome-sequencing/tcga>); BRACS (3-class and 7-class variants) (<https://www.bracs.icar.cnr.it/>); VQA-RAD (<https://osf.io/89kps/>); SLAKE (<https://www.med-vqa.com/slake/>); TCGA-UT (<https://huggingface.co/datasets/dakomura/tcga-ut>). Unless otherwise specified by the hosting platforms, these datasets are released for academic research use upon acceptance of the corresponding terms of use. Dataset supporting the findings of this study are available from the aforementioned repositories. Source data are provided with this paper.

Code Availability

The pretrained models, as well as source code for training and inference, can be accessed at <https://github.com/wangshansong1/MMKD-CLIP>.

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Author Contributions

S.W. conceived and designed the study. S.W., Z.J., M.H., M.S., Y.G., and F.Z. acquired the public data. S.W., C.-W.C., J.R., D.S.Y., and R.L.Q. performed statistical analyses. S.W. and M.S. trained and validated the deep learning network. S.W., J.E.B., X.Y. and Y.G. implemented quality control of data and algorithms. S.W., M.S., and X.Y. prepared the first draft of the manuscript. All the authors contributed to the preparation of the manuscript and approved its submission for publication.

Competing Interests

The authors declare no competing interests.

Figure Legends/Captions

Figure 1. Overview of MMKD-CLIP dataset curation, model training, and evaluation. a. Dataset assembly for pretraining and distillation. We extracted over 2.9 million paired image-text data from the PMC-OA database. Nine pretrained medical CLIP teacher models including BMCA, PMCCCLIP, GenMedClip, MedCLIP, BiomedCLIP, UniMedCLIP, PLIP, PUBMEDCLIP, and QuiltNet supplied over 16.5 million image and text feature vectors for student distillation. b. Pretraining of MMKD-CLIP based on contrastive learning. Each raw image-text pair is encoded via separate image and text encoder. Representations are aligned by standard contrastive learning to establish an initial multimodal embedding space. c. Knowledge distillation pipeline. The student model is trained using both feature distillation (FD) loss and interactive contrastive learning (ICL) loss. d. General and clinical applications. Post-training, MMKD-CLIP supports (1) zero-shot image classification, (2) linear probing image classification, (3) text-to-image/image-to-text retrieval, (4) visual question answering and (5) outcome prediction. e. Evaluation dataset distribution and downstream performance. Concentric rings depict modality coverage across 58 benchmark datasets spanning CT, MRI, ultrasound, pathology, dermatology, fundus, OCT, endoscopy, radiography and X-ray. Radar chart and bar charts compare MMKD-CLIP against state-of-the-art vision-language models on different metrics.

Figure 2. Zero-shot classification performance of MMKD-CLIP across nine medical imaging modalities. Bar charts show the mean AUC (%) for each modality (MRI, endoscopy, fundus, OCT, ultrasound, dermatology, CT, X-ray, and pathology), where bar height represents the mean across datasets within that modality. Dots indicate the AUC of each individual dataset, with each dataset treated as one experimental unit (n = number of datasets in that modality). Error bars indicate 95% confidence intervals at the modality level, estimated from 1,000 paired bootstrap replicates of the modality-level mean AUC. For each bootstrap iteration, AUC values were resampled with replacement within each dataset and then averaged across all datasets within the modality. Statistical significance was assessed using two-sided empirical paired bootstrap tests comparing MMKD-CLIP with the top three competing models for each modality, based on the bootstrap distribution of differences in modality-level mean AUC. Source data are provided as a Source Data file.

Figure 3. Linear probing image classification and cross-modal retrieval performance of MMKD-CLIP. a. Linear probing image classification. For each of nine imaging modalities (X-ray, fundus, MRI, OCT, pathology, endoscopy, CT, dermatology, ultrasound), we trained a single linear classifier on 1%, 10% and 100% subsets of the respective training data. Lines show mean AUC (%) across datasets within each modality. MMKD-CLIP is highlighted in red, and nine teacher models are shown in other shades. b. Cross-modal retrieval. Left panels report image-to-text (I2T) and text-to-image (T2I) Recall@1, Recall@10 and Recall@50 on the MedTrinity-25M subset ($n = 3,771$ queries). Right panels show the same metrics on the BookSet benchmark ($n = 4,265$ queries). The unit of analysis was the individual query. Error bars indicate 95% confidence intervals estimated using 1,000 nonparametric bootstrap resamples with replacement over queries, using the full set of queries in each resample. Statistical significance was assessed using two-sided empirical paired bootstrap tests over queries, comparing MMKD-CLIP with the corresponding best comparator model for each retrieval metric on each benchmark. Source data are provided as a Source Data file.

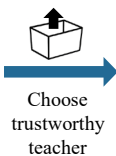
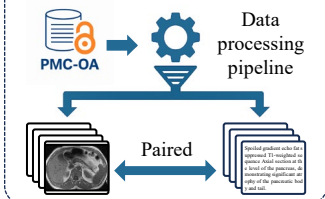
Figure 4. Visual question answering performance and qualitative examples. a. VQA accuracy (%) on the SLAKE and VQA-RAD benchmarks. Bars show mean accuracy and 95% confidence intervals for the “Open”, “Closed”, and “Overall” settings in each dataset. The unit of analysis was the individual question, and each question-answer pair was treated as one independent data point ($n = 602$ open, $n = 430$ closed, and $n = 1,032$ overall question-answer pairs for SLAKE; $n = 145$ open, $n = 299$ closed, and $n = 443$ overall question-answer pairs for VQA-RAD). Error bars indicate 95% confidence intervals estimated using 1,000 bootstrap resamples with replacement over question-answer pairs. Statistical significance versus the next-best model was assessed using two-sided empirical paired bootstrap tests over question-answer pairs, using the same resampled question-answer pairs for both models at each bootstrap iteration. Sub-figure b and sub-figure c qualitative examples and model responses on SLAKE and VQA-RAD. Five representative prompts (columns) spanning anatomical, pathological and spatial questions are shown with the correct answer and answer type (Open/Closed) beneath each image. Model first predicts the type of question (Open or Closed) and then gives the correct answer. For each model, green checks denote correct classification and content, and red crosses denote errors in either component. MMKD-CLIP achieves the highest consistency across both open- and closed-format queries. Source data are provided as a Source Data file.

Figure 5. Survival prediction with MMKD-CLIP and PLIP. a. Kaplan-Meier curves for risk stratification across TCGA cohorts. For 12 cancer types (BRCA, LUAD, LUSC, KIRC, HNSC, BLCA, STAD, SARC, CESC, PAAD, UCEC, MESO), patients were split into high-risk (red) and low-risk (blue) groups based on median risk scores derived from whole-slide images and clinical reports. Survival probability is plotted over 20 years; hazard ratios (HR) and two-sided log-rank p-values are indicated on each panel. b. C-Index performance for survival prediction. Bars represent the mean C-Index across multiple cross-validation folds, with error bars indicating the standard deviation (s.d.) derived from five-fold experiments. Three input modalities were evaluated: image-only (cyan), report-only (pink), and combined image+report (apricot). The upper and lower panels correspond to MMKD-CLIP and PLIP foundational models, respectively. For the overall evaluation, $n = 12$ independent TCGA datasets serve as the primary units of analysis. Statistical significance on the overall C-Index was assessed using a two-sided Wilcoxon signed-rank test to account for parity across shared data partitions between models. Source data are provided as a Source Data file.

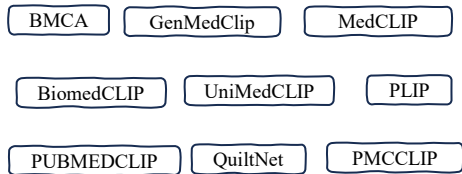
Figure 6. Supervised cancer diagnosis performance and ablation study of MMKD-CLIP. a. Supervised fine-tuning results on cancer diagnosis. Bar plots show AUC (%) for each model on four histopathology cohorts (BRACS-3, BRACS-7, TCGA-NSCLC, and Camelyon16) after fine-tuning. Error bars indicate 95% confidence intervals estimated using 1,000 paired bootstrap resamples of the test set for each cohort. Statistical significance versus the next-best model within each cohort was assessed using two-sided empirical P values derived from the paired bootstrap distribution of AUC differences. b. Ablation study on zero-shot image classification. We compared (1) contrastive pretraining only, (2) MMKD-CLIP without FD loss, (3) MMKD-CLIP without ICL loss, and (4) the full MMKD-CLIP. Bars indicate mean AUC (%) across datasets for the overall benchmark and for modality-specific subsets (X-ray, pathology, CT, and MRI), where n denotes the number of datasets in each group. Error bars represent 95% confidence intervals estimated using 1,000 paired bootstrap resamples across datasets. At each bootstrap iteration, dataset-level AUC values were averaged within each group to obtain 1,000 bootstrap replicates of the mean AUC for each method. Two-sided empirical P values were computed from the paired bootstrap distribution of differences between methods. Source data are provided as a Source Data file.

a. Dataset for pretraining and distilling MMKD-CLIP

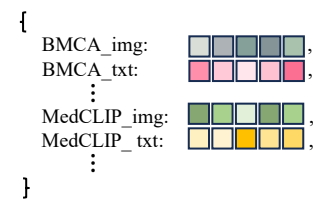
2.9M PMC data for pretraining



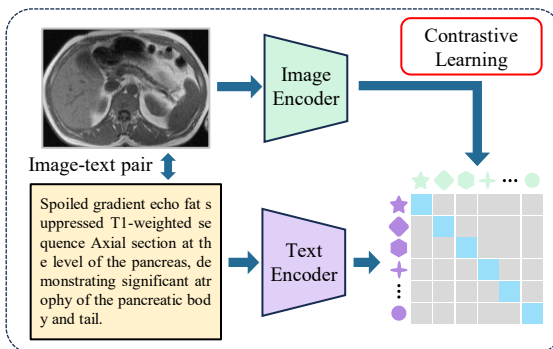
Bag of pretrained medical CLIP



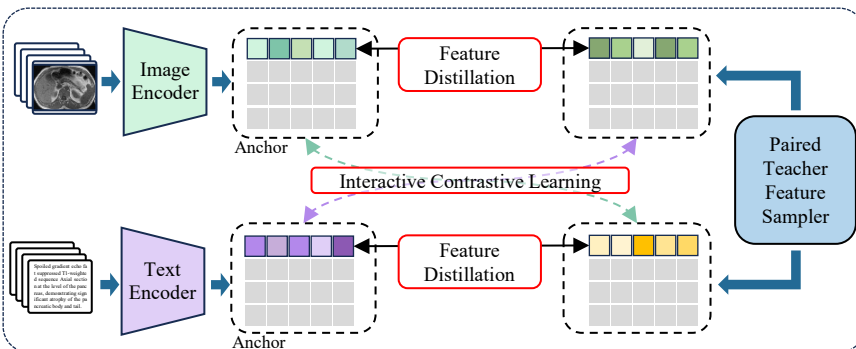
16.5M CLIP feature for distilling



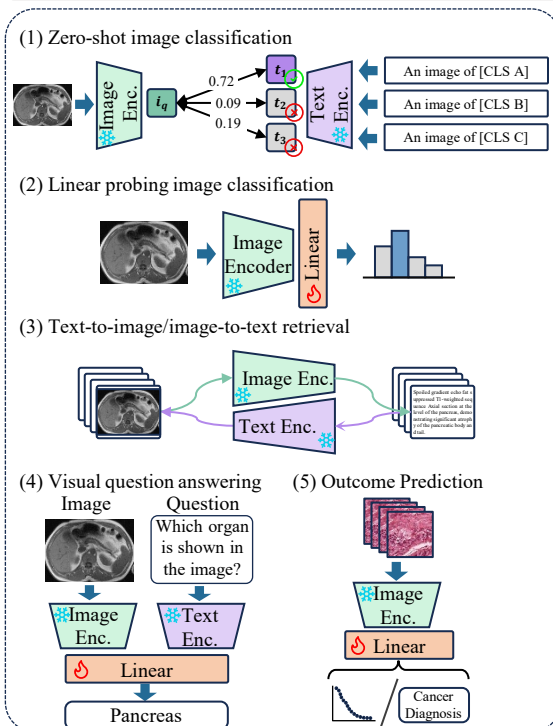
b. Pretraining of MMKD-CLIP



c. Distillation of MMKD-CLIP

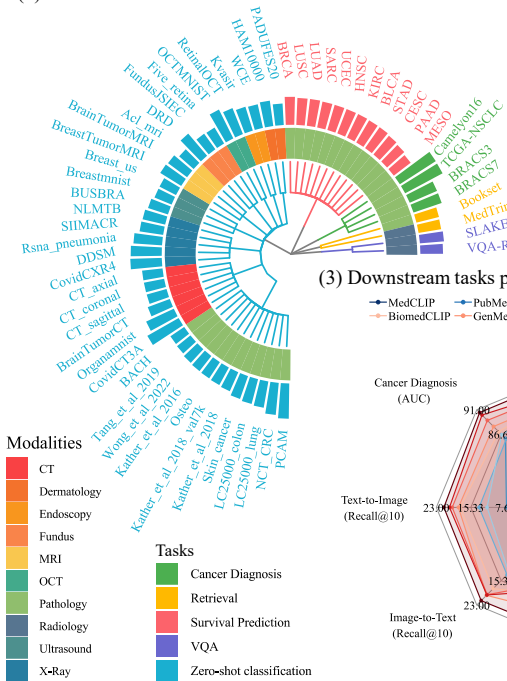


d. General-purpose clinical applications

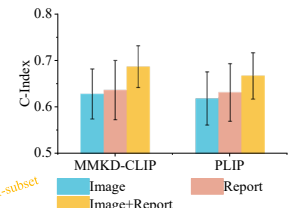


e. Dataset statistics and performance

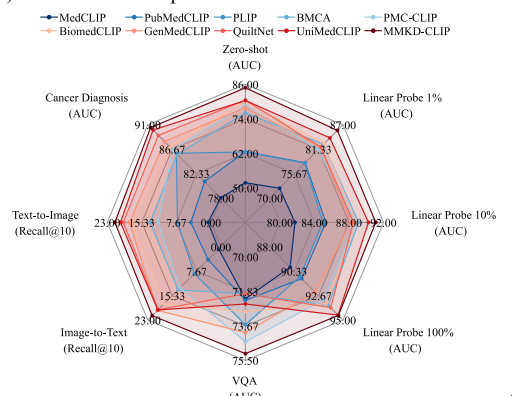
(1) Evaluation dataset distribution

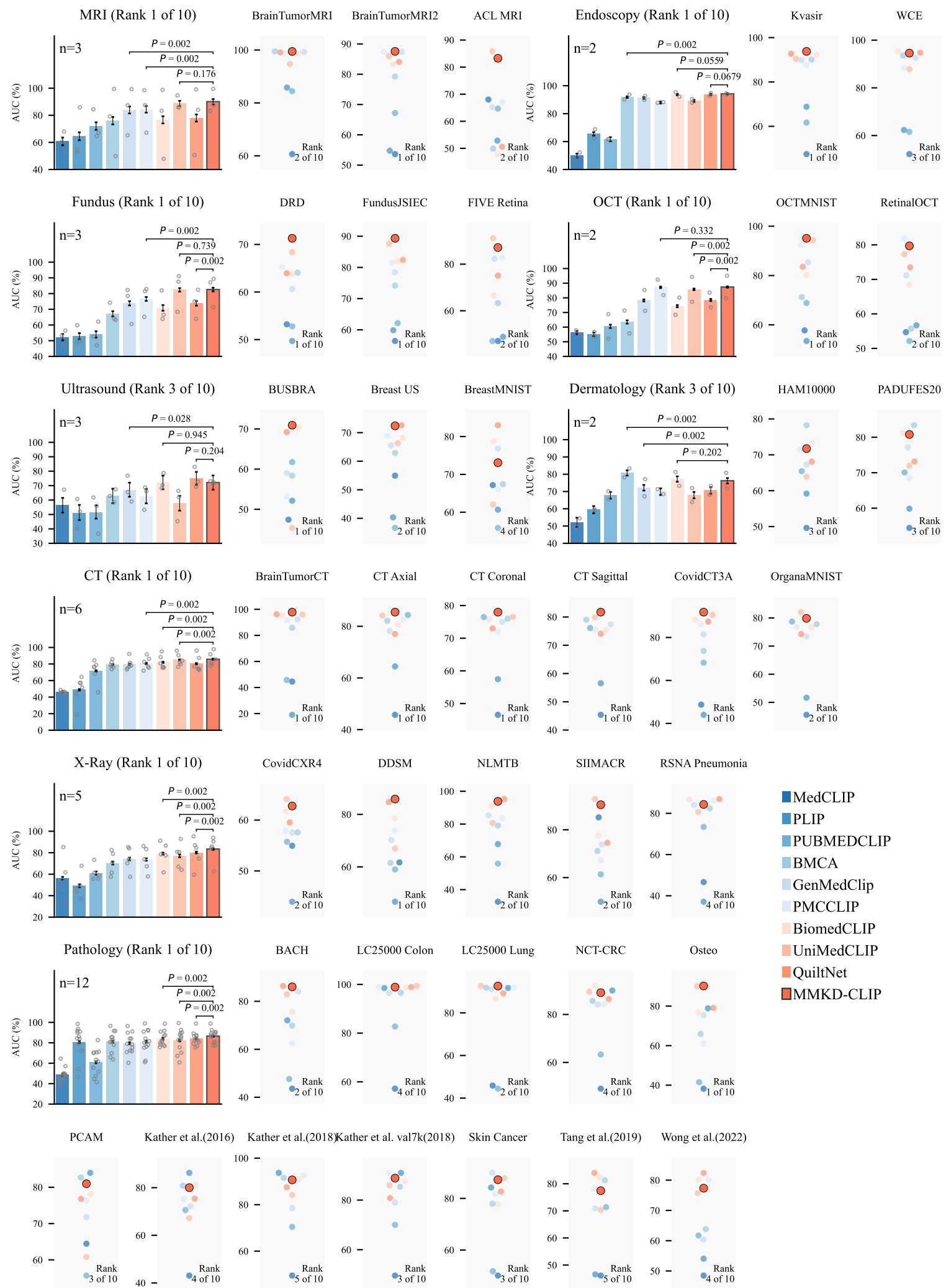


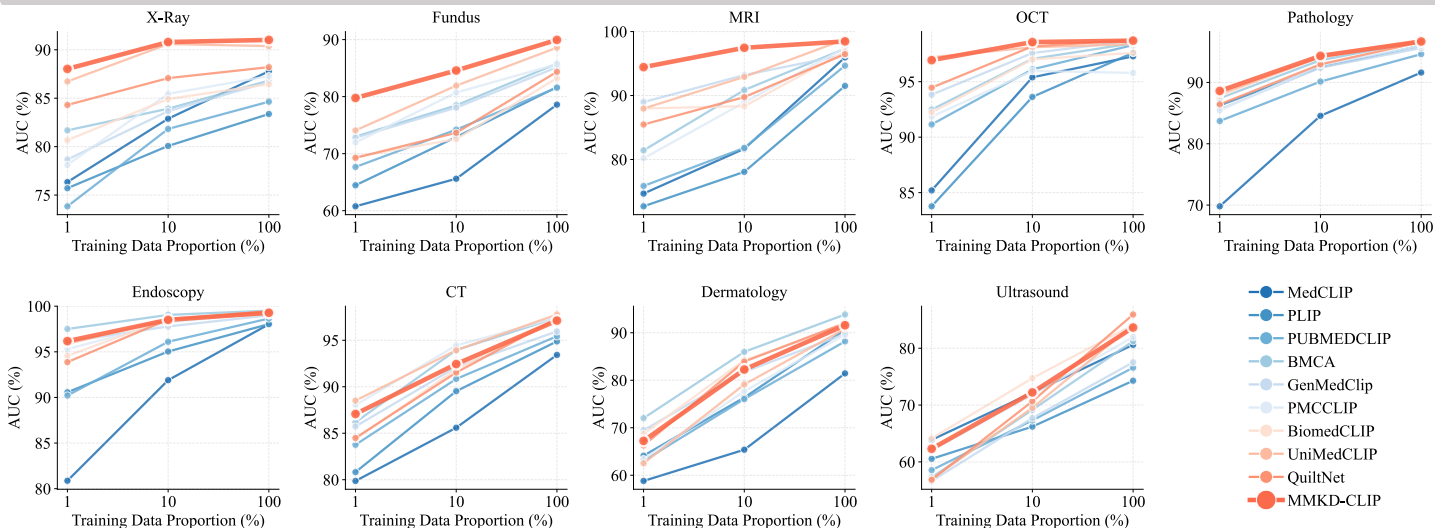
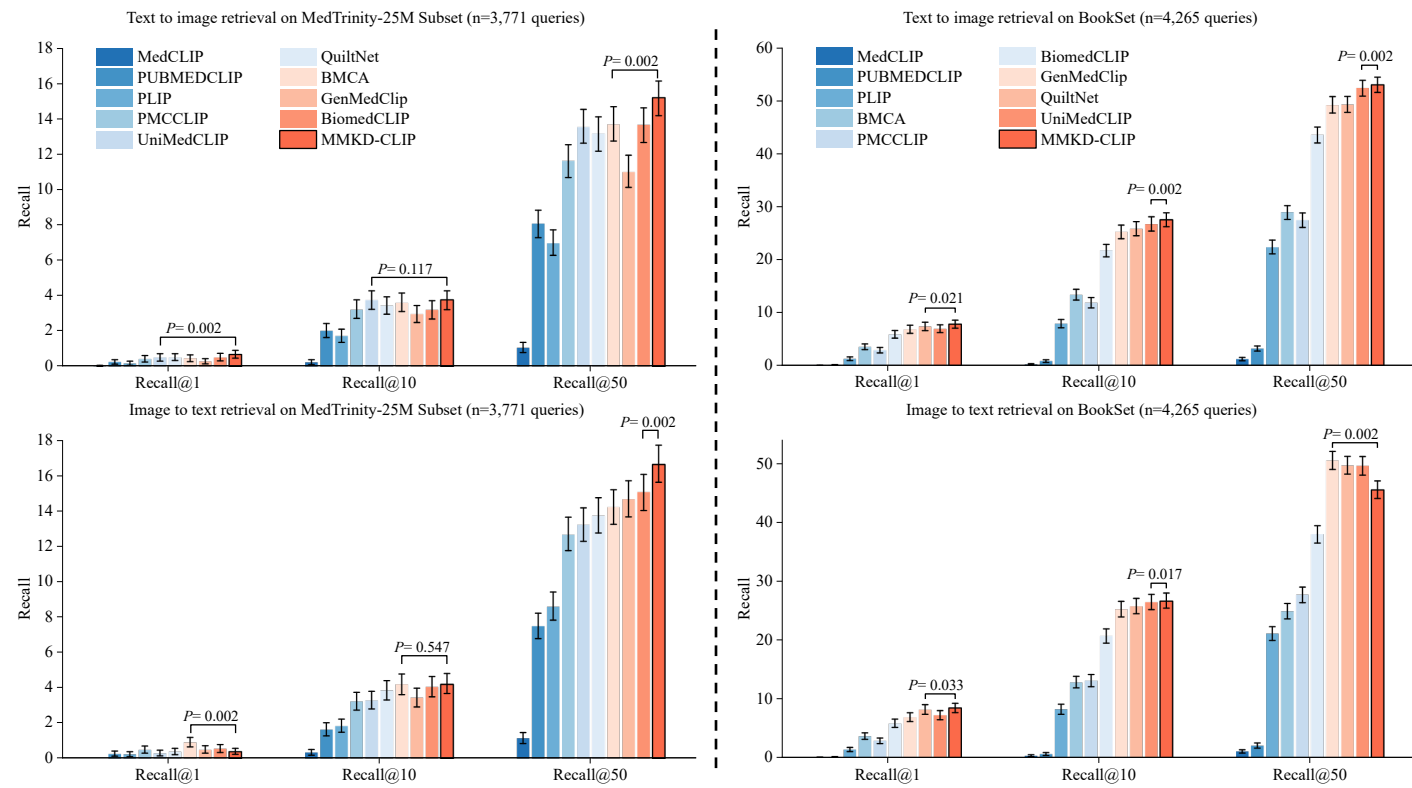
(2) Survival prediction performance



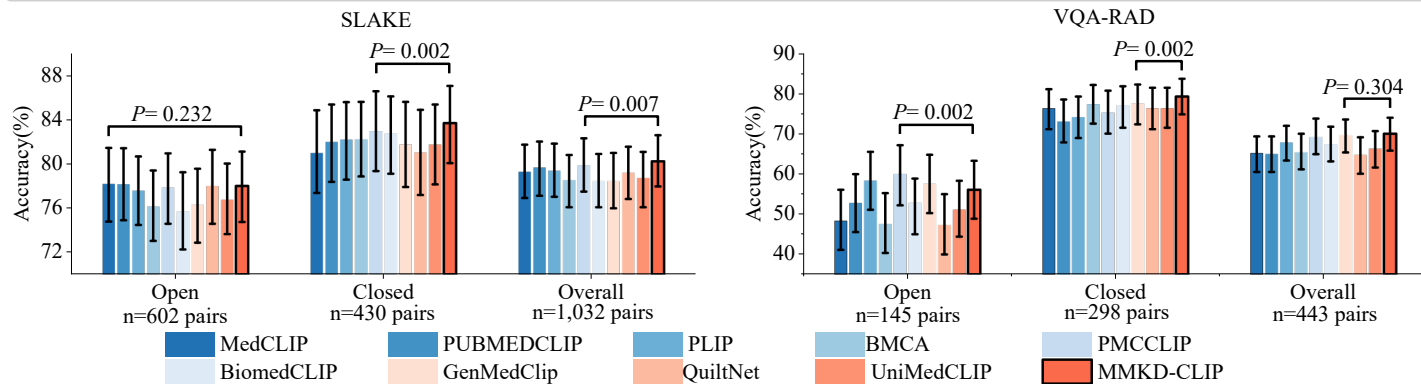
(3) Downstream tasks performance





a. Linear probing image classification**b. Cross-modal retrieval**

a. VQA performance



b. Visualization in SLAKE



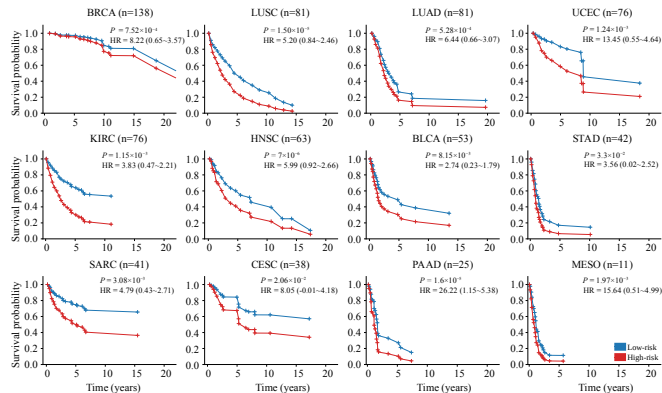
Question:	How many lungs have existed in this image?	Is this a coronal section?	What organ is the gray part on the right of the image?	What is the organ on the left (patient's right) side of this image?	Where is the atelectasis in this image?
Answer/Type:	2/Open	No/Closed	Small Bowel/Open	Liver/Open	Lower Right Lung/Open
MMKD-CLIP:	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓
BiomedCLIP:	✗ / ✓	✓ / ✓	✗ / ✓	✗ / ✓	✗ / ✗
BMCA:	✓ / ✓	✓ / ✓	✓ / ✓	✗ / ✗	✗ / ✓
GenMedClip:	✗ / ✓	✗ / ✗	✗ / ✓	✗ / ✗	✓ / ✓
MedCLIP:	✓ / ✓	✗ / ✗	✗ / ✓	✗ / ✗	✗ / ✗
PLIP:	✓ / ✓	✗ / ✗	✓ / ✓	✓ / ✓	✗ / ✓
PMCCCLIP:	✗ / ✓	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓
PUBMEDCLIP:	✗ / ✗	✗ / ✗	✓ / ✓	✓ / ✓	✓ / ✓
QuiltNet:	✓ / ✓	✓ / ✓	✗ / ✓	✗ / ✓	✗ / ✗
UniMedCLIP:	✓ / ✓	✓ / ✓	✗ / ✓	✓ / ✓	✓ / ✓

c. Visualization in VQA-RAD

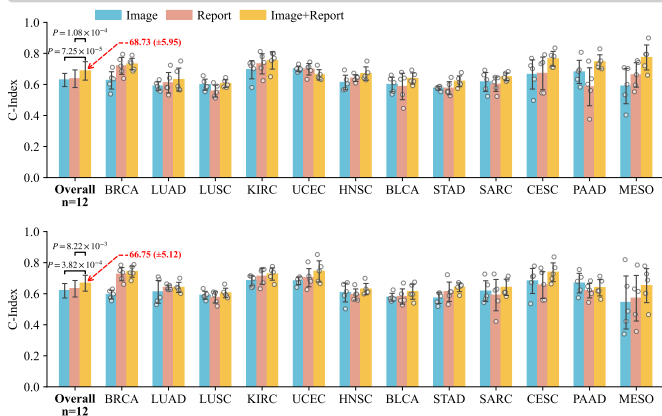


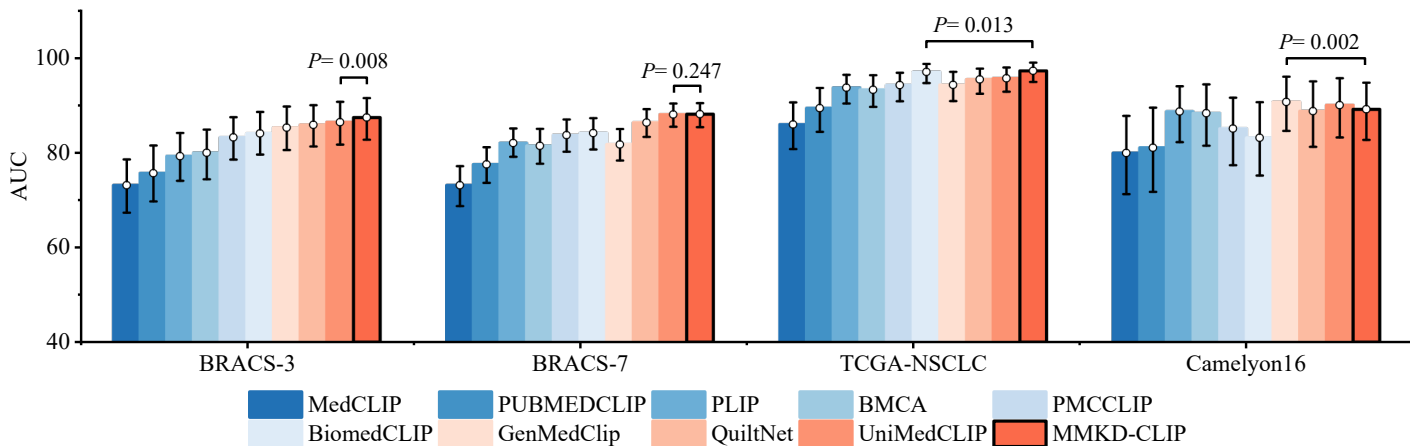
Question:	Is the skull fractured?	What are the hyper-dense lesions noted at the edges of the aorta?	Where is the lesion located?	What is in the left apex?	Do the sulci look abnormal in this image?
Answer/Type:	No/Closed	Calcified atherosclerosis /Open	Anterior mediastinum /Open	A bullous lesion /Open	No/Closed
MMKD-CLIP:	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓
BiomedCLIP:	✗ / ✗	✗ / ✓	✗ / ✗	✗ / ✓	✓ / ✓
BMCA:	✓ / ✓	✗ / ✗	✗ / ✓	✗ / ✓	✗ / ✗
GenMedClip:	✓ / ✓	✓ / ✓	✗ / ✗	✗ / ✓	✓ / ✓
MedCLIP:	✓ / ✓	✓ / ✓	✗ / ✓	✓ / ✓	✗ / ✓
PLIP:	✓ / ✓	✗ / ✓	✗ / ✓	✓ / ✓	✓ / ✓
PMCCCLIP:	✗ / ✗	✗ / ✗	✗ / ✓	✗ / ✓	✓ / ✓
PUBMEDCLIP:	✓ / ✓	✗ / ✗	✓ / ✓	✓ / ✓	✗ / ✓
QuiltNet:	✗ / ✗	✗ / ✓	✗ / ✗	✗ / ✓	✓ / ✓
UniMedCLIP:	✓ / ✓	✗ / ✓	✗ / ✓	✗ / ✓	✗ / ✗

a. Survival probability visualization



b. Survival prediction performance on MMKD-CLIP and PLIP



a. Supervised cancer diagnosis result**b. Ablation study based on zero-shot image classification**